



ELEE 4230/6230: Sensors and Transducers

Course Project - Summer Semester 2026

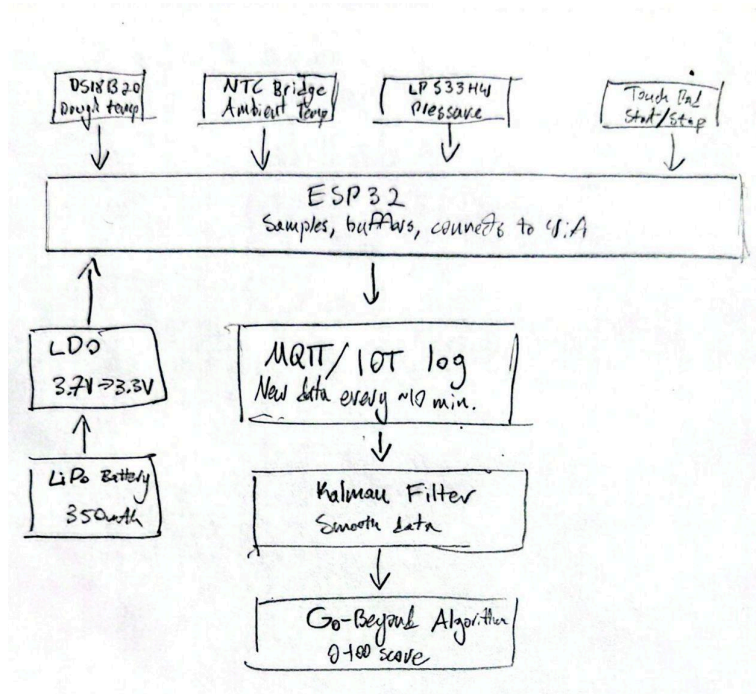
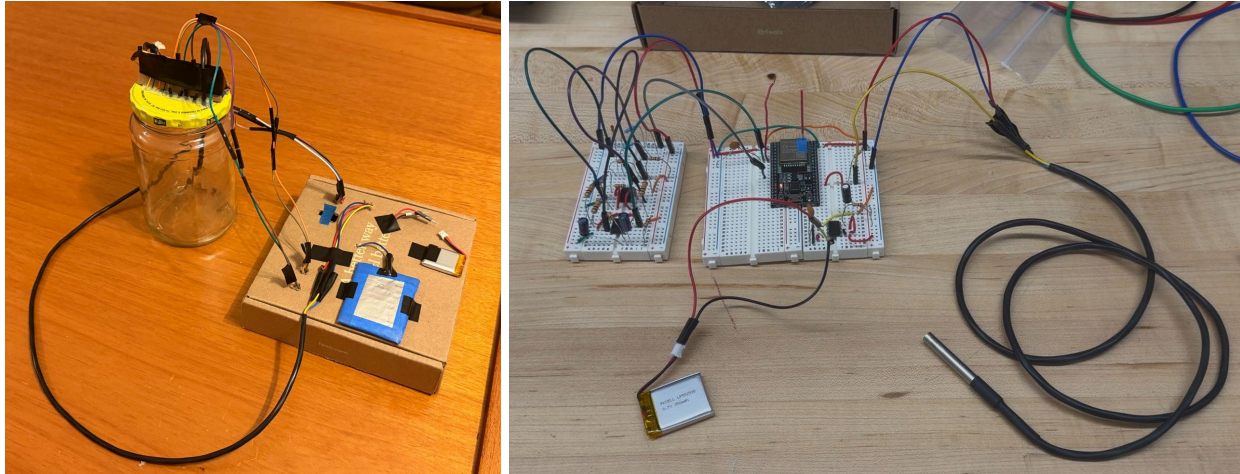
Bread Fermentation Sensor System

Gibson Sparks

[Table of Contents](#)

0. A Tour of the Fermentation Sensor System.....	2
1. Problem Definition, Environment, and Project Constraints.....	4
2. Sensor Selection and Design Decisions.....	7
3. RAW Sensor Dynamics and Signal Processing.....	9
Raw Sensor Physics and Modeling.....	9
Signal Processing Mountain.....	9
Final Implementation (Circuit and Power).....	10
Filtering and Kalman Implementation.....	11
4. Capacitive Sensor (User Input System).....	13
Physics and Construction.....	13
Signal Processing and Implementation.....	13
System Integration.....	14
Limitations and Performance.....	15
5. Other Sensors Used.....	16
6. Digital Logic and Algorithms.....	18
7. Final Design and Experimental Results.....	20
8. Failure Analysis.....	23
9. Physical Implementation.....	25
10. Course Reflection.....	27
Project Reflections.....	27
Lecture Reflections.....	27
11. References.....	28
Appendix.....	29

0. A Tour of the Fermentation Sensor System



Operation

The system is powered from an LiPo battery which is stepped down to 3.3V through an LDO. A finger on the capacitive pad starts a session. Every 20 seconds, the ESP32 reads all three sensing channels: dough temperature from the DS18B20, ambient temperature from the thermistor bridge, and headspace pressure from the LPS33HW, which are buffered in RAM. Roughly every 10 minutes, it connects to WiFi, publishes the buffered batch over MQTT to a logging client, and disconnects to save power and reduce system-voltage noise. At the end of a session, the data is sent through a Kalman filtering algorithm to reduce noise. After that, the data is sent to a Go-Beyond algorithm, which fits a power-law curve to the pressure data. This algorithm also corrects for starting dough temperature and yeast mass, but ends in producing a single 0-100 fermentation score.

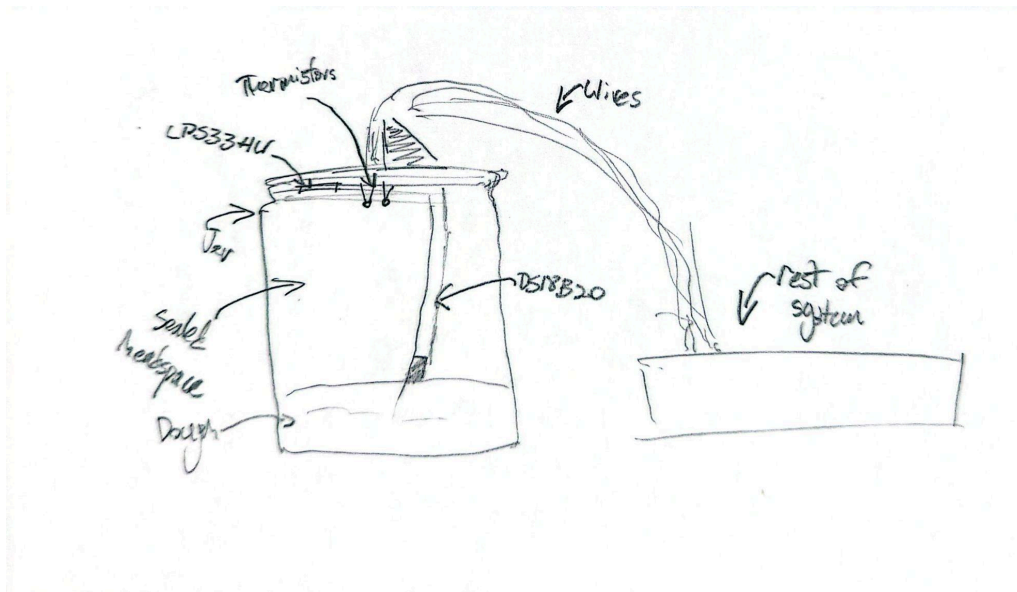
1. Problem Definition, Environment, and Project Constraints

Problem Definition

Home bakers currently judge bread dough fermentation readiness by look, feel, and elapsed time, methods that don't quantify the process, don't transfer between recipes or ambient conditions, and give no early warning of over- or under-proofing. The problem this project addresses is: how can fermentation progress inside a sealed dough vessel be measured quantitatively and non-destructively, producing a single, comparable measure of "how far along" the process is, without opening the vessel or disturbing the dough?

This shows up in two related ways. At the level of a single bake, temperature and yeast health both affect how far a dough develops in a given proofing window, and elapsed time alone can't tell them apart; a baker following a recipe's "two-hour proof" has no way to know whether a flat loaf came from half the yeast being non-viable, an ambient temperature issue, or something else. With a quantitative pressure-based reading, they could instead see empirically that fermentation is running behind schedule and know to extend the proof until the reading reaches its normal range, rather than guessing.

At the level of comparing across bakes, that same lack of a common yardstick makes it hard to keep batches consistent while developing or scaling a recipe. Because the underlying measurement, headspace pressure rise, means the same thing regardless of which batch it comes from, a small portion of any batch placed into the sensor vessel gives a reading that extrapolates back to the whole batch, giving a baker one consistent, quantifiable number to compare across attempts instead of relying on memory or feel. Both uses ultimately rest on the same capability: turning fermentation progress into one number that means the same thing every time it's measured.



Environment

The measurement environment is the headspace above a rising dough inside a sealed vessel. Conditions include:

- Temperature ranging from a freshly-mixed dough's initial $\sim 30\text{-}35^\circ\text{C}$ down toward room temperature ($\sim 20\text{-}26^\circ\text{C}$) as the batch equilibrates.

- Rising CO₂ concentration and headspace pressure as fermentation proceeds; well above ambient CO₂ levels but still low absolute pressure.
- High local humidity within the container. Electronics within the vessel need to be protected.
- Session durations on the order of 30 minutes to 3 hours.

Safety Checklist

The sealed proofing chamber reaches near-saturation humidity (90-100% RH) and direct contact with moist dough; safety is maintained by using waterproof or gel-encapsulated housings, sealing all container wire pass-throughs with hot glue and silicone. Minor absolute pressure fluctuations inside the container and exposure to mild fermentation byproducts (such as trace carbon dioxide and ethanol vapor) present negligible hazard. Power safety is established using a 3.3V low-dropout (LDO) regulator that maintains a steady voltage output as the battery discharges. Additionally, low-pass filters and digital signal formatting help insulate the measurement lines against electrical noise and radio frequency interference from nearby RF signals.

Constraints

Constraints	Value / Range	Source
Dough temperature range	~20-40 °C	CP1 vessel/recipe assumptions; observed reference-bake data
Ambient/headspace temperature range	~20-30 °C	This value was assumed based on likely dough temperatures, however a sensor was chosen that could be adjusted to fit a +40 °C range.
Headspace pressure range	Could be adjusted based on container size.	Sensor max 1260 hPa; observed working range ~989-1000 hPa
Temperature accuracy target	<0.1 °C	Gives at least 100 unique steps for the 10 °C range expected.
High humidity/condensation	90-100% RH	The wet dough will saturate the surrounding environment. Upon the dough's cooling, condensation is likely to form.
Power budget	>4 hours of usage	Dough proof would at most last 3 hours. Longer the better.

Limitations

One main limitation of this system, when used to find consistency among large batches, is that the small subsample placed in the vessel won't experience exactly the same conditions as the rest of the dough over the same period. A given mass of dough has a fixed

volume-to-surface-area ratio that changes the moment a smaller piece is separated into its own vessel: the smaller sample and its headspace equilibrate to ambient temperature faster than a much larger bulk of dough would, so the subsample can reach a given reading at a systematically different point in the timeline than the main batch does. Mixing distribution is a second source of mismatch; if yeast, sugar, or starter aren't perfectly homogeneous across a mixed batch, the small sample used for measurement carries whatever local variation ended up in that particular scoop, rather than a true batch average. None of this makes the reading meaningless, but it's best treated as a comparative signal between samples prepared the same way, rather than an absolute stand-in for the full batch.

2. Sensor Selection and Design Decisions

Comparison Table

The most open sensor decision in the project was the third channel (beyond the two temperature sensors), where three options were seriously evaluated:

Option	Cost / Complexity	How does it fit?	Key drawback
Strain gauge (weight)	\$30+, Needs 2 gauges + 2 ADCs to cancel drift.	Unsealed production loses mass due to sugar conversion into CO ₂ . The gauges could measure this change.	Creep, vibration, and thermal drift over a 1-3 hr session make single-gauge error too large; a second gauge for comparison roughly doubles cost, but error is likely too great.
NDIR CO ₂ sensor	~\$10, simple I2C interface	Fermentation produces a lot of CO ₂ that could be measured to track the process.	Sealed-vessel CO ₂ concentrations run far above the hundreds-to-low-thousands ppm range these sensors are built for.
Absolute pressure sensor (LPS33HW, selected)	~\$10, I2C, water-resistant sensing element	Headspace pressure rises quantitatively and directly with CO ₂ production.	The breakout board (not the sensor die) needs protection from humidity.

Final Sensor Selection

The LPS33HW pressure sensor was chosen because CO₂ buildup in a sealed vessel relates directly and quantitatively to fermentation activity, exactly the signal the project needs, while avoiding the strain gauge's drift/cost problems and the NDIR sensor's mismatched concentration range. For internal dough temperature, a pre-packaged DS18B20 (stainless steel, waterproof, digital) was chosen since it gave a simple, already-calibrated internal reading. A set of thermistors were chosen to give the DS18B20 a redundant temperature sensing mechanism, and because they are robust, and offer a non-linear transfer function. The system therefore ended up with one custom RAW sensor (thermistor bridge) plus two pre-packaged sensors (DS18B20, LPS33HW).

Design Decision Blocks

Decision 1: Opposing half-Wheatstone bridge for the ambient temperature channel.

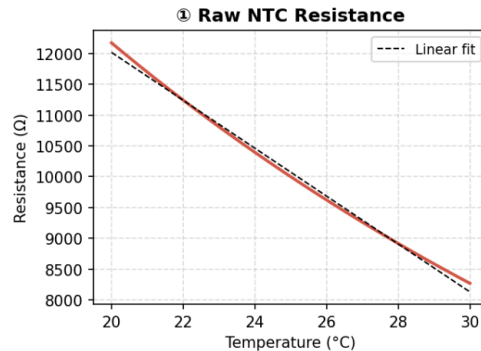
A Wheatstone bridge was used for two main reasons. The first is that it removes noise affecting both sensors, and results stay steady even if the input power supply voltage shifts slightly. Additionally, the bridge acts to centralize the non-linearity near 25 °C, which makes the data more accurate near where most data is taken.

Decision 2: LDO regulator instead of a switching regulator.

Many other systems were considered, such as a switching buck regulator, which is more familiar. However, given the battery, the buck converter likely wouldn't have worked as well because the battery-to-rail voltage differential is small enough that the LDO's efficiency loss is acceptable, and a switcher's inherent switching noise would couple into the sensitive thermistor-bridge analog front end. The added system noise for a marginal efficiency gain in this specific low-current application wasn't the best decision.

3. RAW Sensor Dynamics and Signal Processing

Raw Sensor Physics and Modeling



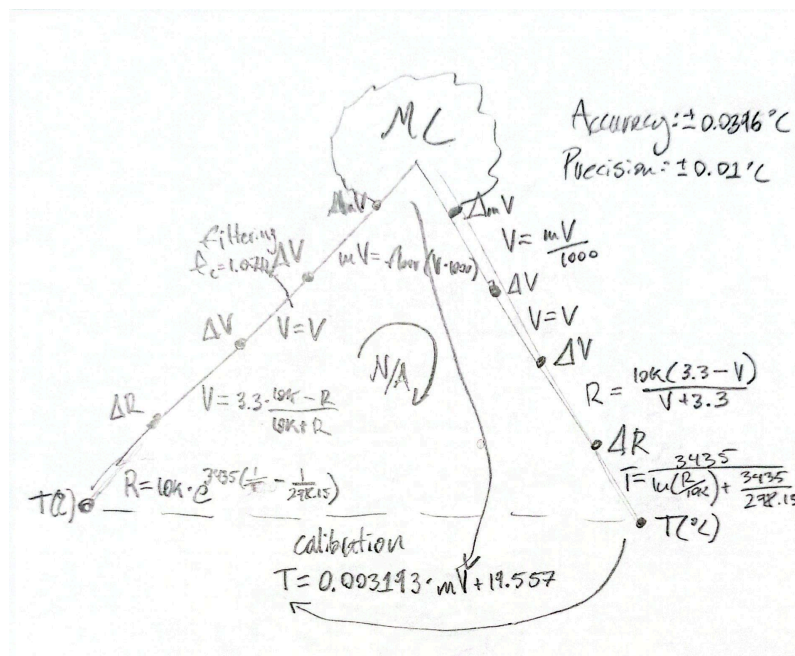
NTC thermistor resistance is governed by the B-parameter equation:

$$R(T) = R_0 \cdot e^{B \left(\frac{1}{T} - \frac{1}{T_0} \right)}$$

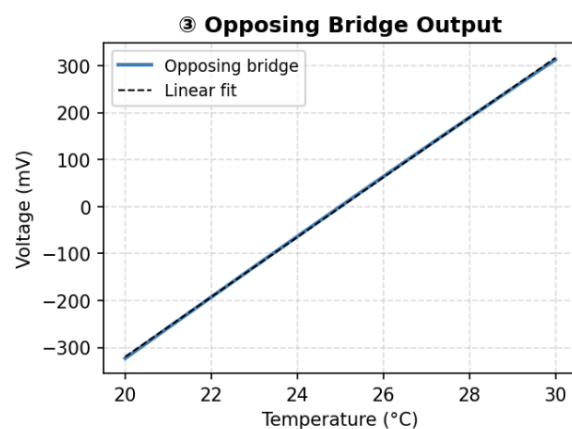
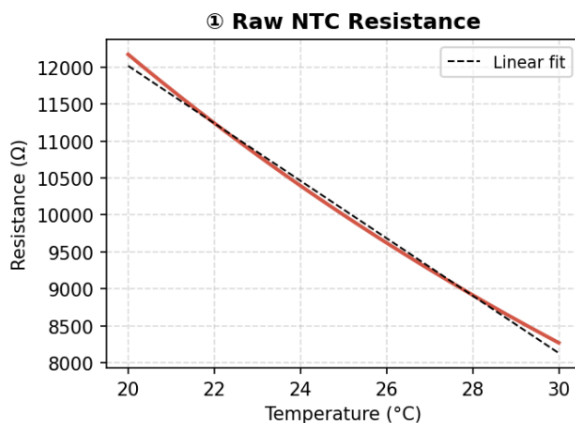
with $R_0 = 10 \text{ k}\Omega$ at $T_0 = 298.15 \text{ K}$ (25°C), and $B = 3435 \text{ K}$.

More charge carriers are thermally excited across the semiconductor bandgap as temperature rises, so conductivity increases exponentially. The biggest limitations of this model is that the B-parameter model is itself an approximation. In reality, B varies weakly with temperature, so the fit degrades the further the thermistor gets from 25°C . That is in addition to the variance of resistance between components which further compounds this approximation's uncertainty.

Signal Processing Mountain

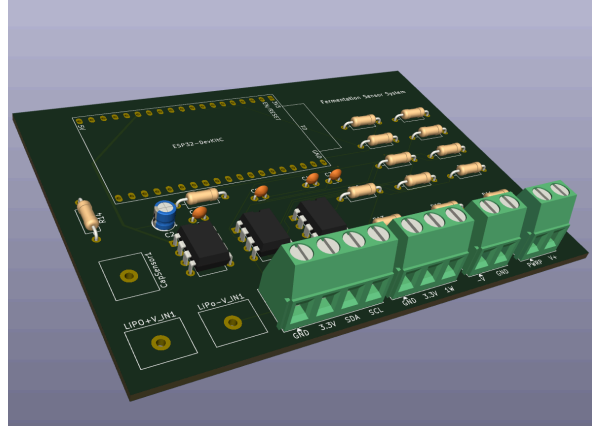
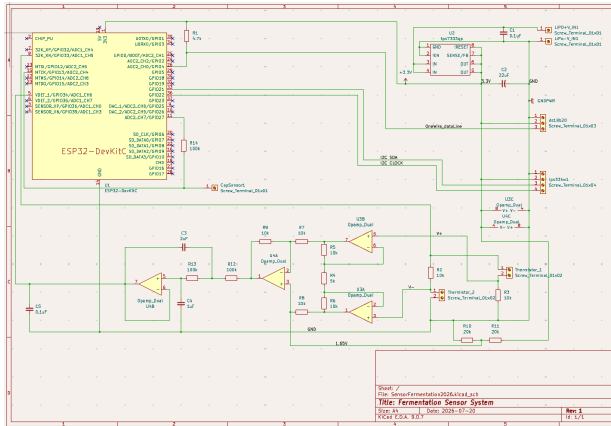


Stage	Transfer relationship	Why it's needed
Physical sensor	$R(T) = R_0 \cdot \exp[B(1/T - 1/T_0)]$	Raw physics
Opposing half-Wheatstone bridge	$V_d = 3.3 \cdot \{(10,000 - R(T)) / (10,000 + R(T))\}$ $V_d = V^+ - V^-$	Cancels even-order nonlinearity and rejects common-mode noise.
Instrumentation amplifier	$V_{out} = 2 \cdot (V^+ - V^-) + 1.65$	Gains the small bridge differential and centers it at $V_{cc}/2$ for the ADC's usable input range (~ 0.1 to ~ 3.2 V)
Sallen-Key low-pass filter	2nd-order Butterworth, $R = 100 \text{ k}\Omega (\times 2)$, $C_1 = 1 \text{ }\mu\text{F}$, $C_2 = 2.2 \text{ }\mu\text{F}$ $f_c = 1.07 \text{ Hz}$	Rejects noise above $\sim 1.07 \text{ Hz}$, since dough temperature changes are inherently slow-moving
ESP32 ADC	$\text{mV} = \text{floor}(V_{out} * 1000)$	Represents the quantization of the voltage when read by the ADC.
Linear calibration fit	$T = 0.003293 \cdot \text{mV} + 19.557 \text{ (} R^2 \approx 1.000 \text{)}$	Converts a linear, noise-rejected voltage into a calibrated temperature reading



where the Opposing Bridge's line of best fit has an $R^2 \approx 1.000$

Final Implementation (Circuit and Power)



Power Architecture

3.7 V, 350 mAh LiPo → TPS7333QP LDO → regulated 3.3 V rail, feeding the ESP32, op-amps, and both digital sensors. 22 μ F and 0.1 μ F decoupling capacitors sit close to the LDO, forming an RC low-pass that attenuates regulator noise before it reaches the bridge or amplifier stage. All of the sensors are powered from the 3.3V rail, and power returns to the GND rail.

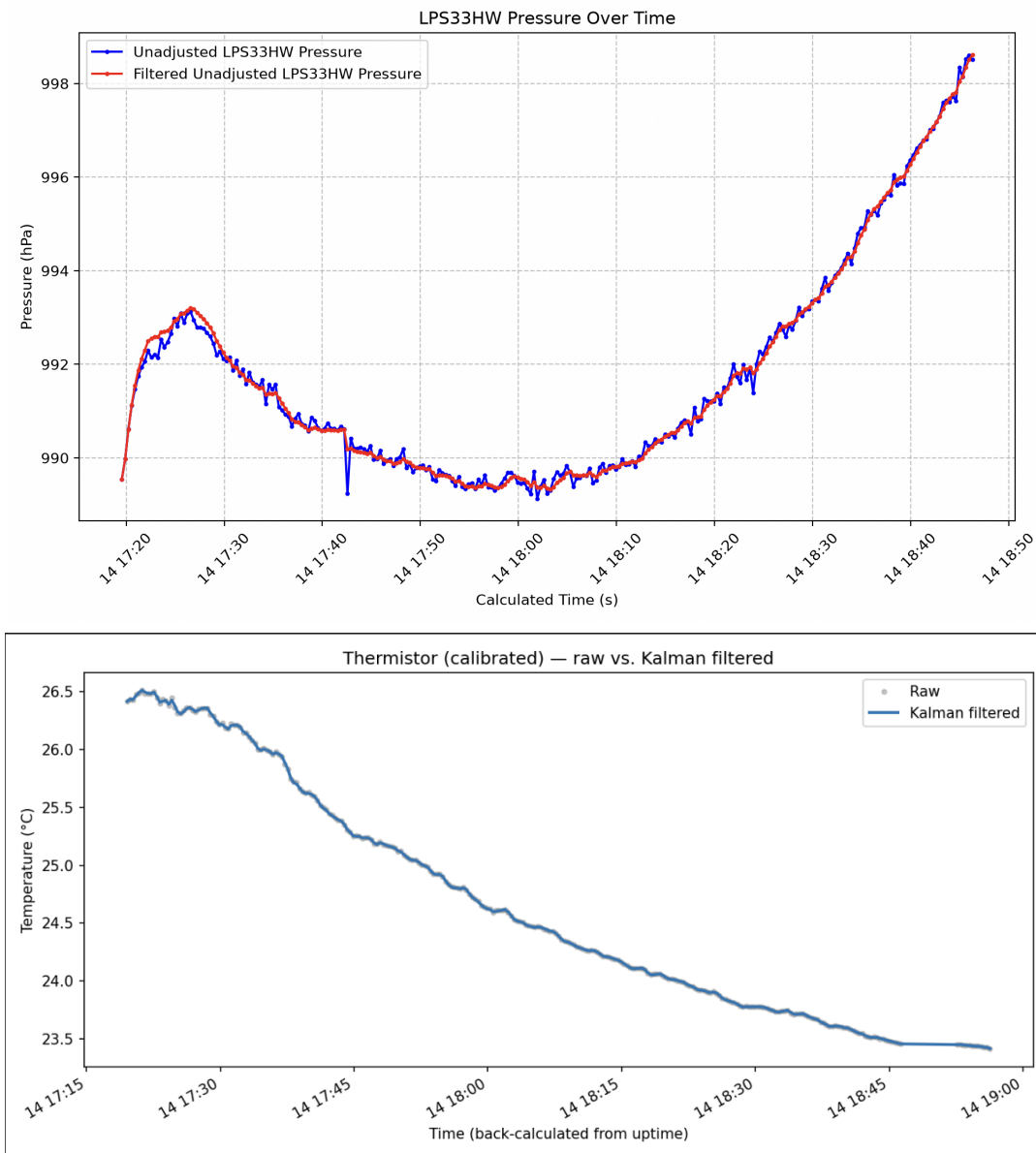
Calibration Method

A multi-point calibration recorded thermistor-bridge output in mV alongside a reference thermometer across the 20-30 °C operating band, producing the linear fit $T = 0.003293 \cdot \text{mV} + 19.557$ used in production.

The same method was used to determine a line of best fit for the DS18B20. Five similarly spaced points were recorded within this range, which were then used to create a line of best fit.

The LPS33HW was also looked at for its accuracy. Tests relating recorded atmospheric pressure at local weather stations appeared to be within 4 hPa of true pressure. Also, pressure change with elevation seemed fairly close to predicted values. Although an error was found, this error was attributed more to the rudimentary methods of taking measurements and not to inaccuracies in the device itself.

Filtering and Kalman Implementation



Kalman Filter Description

Each sensor channel is filtered independently using a constant-velocity Kalman filter. Rather than treating each reading as noise scattered around a fixed value, the filter tracks two things at once: the current value (temperature, in this case) and its rate of change, meaning how fast it's warming or cooling. This lets the filter follow a steady ramp, like a fermentation vessel gradually heating up, instead of just smoothing toward a flat average and lagging behind real trends.

The filter works by alternating between two steps as each new reading comes in. First, it predicts, projecting the value forward using the current rate of change and growing its uncertainty based on how much time has passed since the last reading. Then it updates, pulling that prediction toward the actual new measurement, weighted by how much the filter should trust the new data versus its existing estimate.

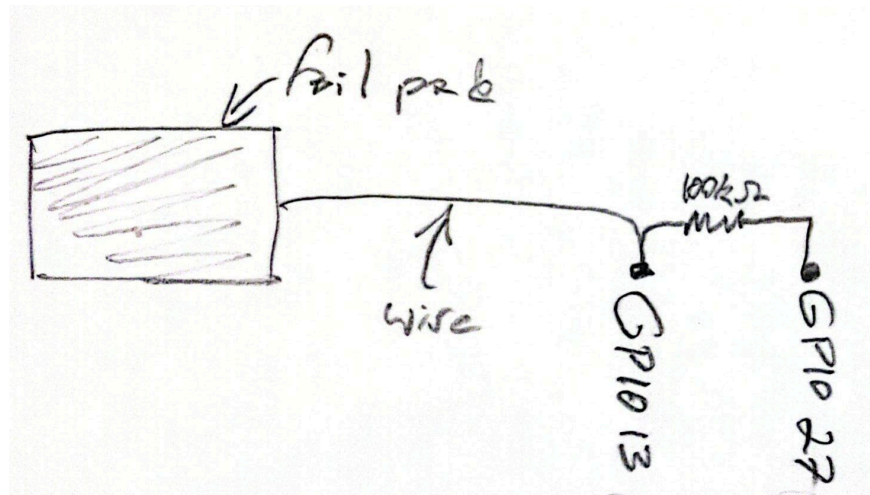
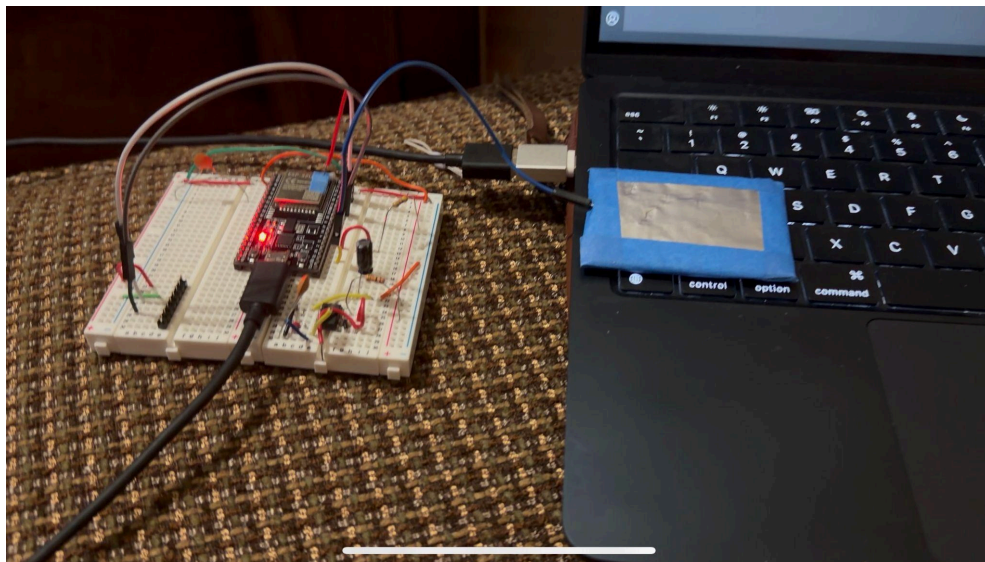
Like the other channels, the thermistor's Kalman filter state is a two-element vector holding the temperature and its rate of change. The measurement noise, R , is auto-estimated from the spread of calibrated thermistor readings within each upload batch, on the assumption that readings taken only seconds apart shouldn't vary much except from sensor noise itself. This can be overridden manually. The process noise, Q , defaults to a conservative $1e-7$, and can be raised if the filtered curve seems to lag behind genuine temperature changes, or lowered if it looks too jumpy.

4. Capacitive Sensor (User Input System)

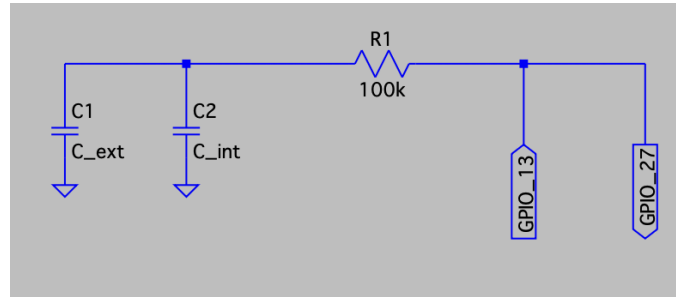
Physics and Construction

The sensor is a 6.6 x 4.85 cm sheet of aluminum foil on a backing, wired from GPIO13 (sense) to GPIO27 (send) through a 100 k Ω resistor. Rather than using the ESP32's dedicated hardware touch pin, the firmware bit-bangs an RC relaxation cycle itself. The sense pin is forced low to discharge, switched to high-impedance input, and the send pin is driven high, letting current charge the sense node through the resistor. A finger brought near the foil adds itself as a parallel-plate capacitor referenced to ground (C_{ext}), sitting in parallel with the node's parasitic capacitance (C_{int}). This slows the RC charging curve, increasing the charging time, and the firmware measures the elapsed time to cross the digital-input logic-high threshold.

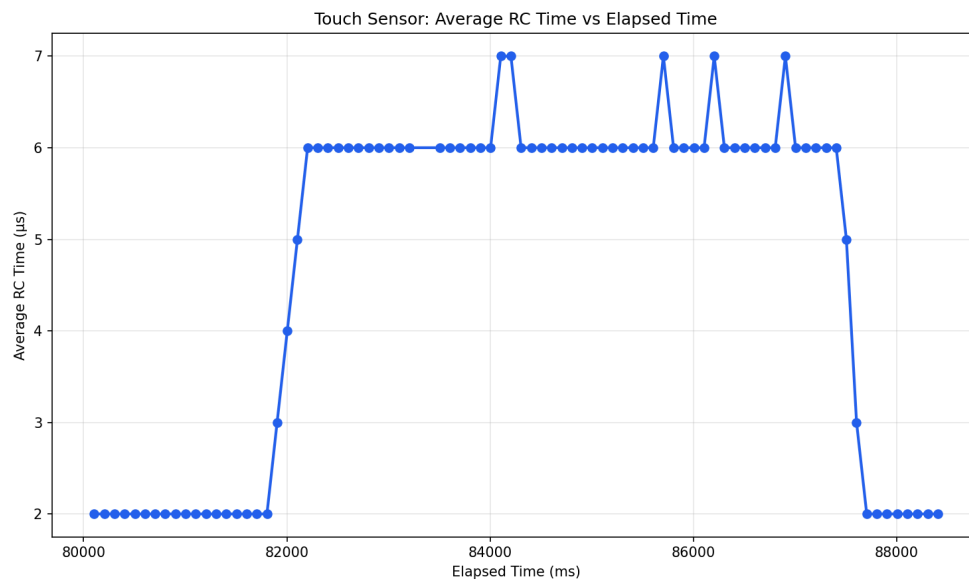
The governing relationship is $t_{\text{elapsed}} = t_{\text{ovh}} + k \cdot C_{\text{ext}}$, where $k = R \cdot \ln(V_{\text{cc}} / (V_{\text{cc}} - V_{\text{th}})) \approx 0.92 \mu\text{s/pF}$. The observed jump from a $\sim 2 \mu\text{s}$ untouched baseline to $6+ \mu\text{s}$ when touched implies roughly 4 pF of added capacitance from a fingertip, consistent with published estimates for human-finger coupling.



Signal Processing and Implementation



The `charge_discharge()` runs once per 100 ms loop cycle and returns the elapsed charge time in microseconds. The system uses a fixed threshold of 6 μs , chosen from the startup baseline (sitting between the $\sim 2 \mu\text{s}$ untouched and $\sim 6+ \mu\text{s}$ touched states). Each threshold crossing toggles a boolean running state variable rather than directly driving an output; this means holding a finger on the pad doesn't repeatedly flip state every 100 ms poll. An averaging scheme is implemented to reduce noise, though in practice it hasn't been strictly necessary.



The high points of the graph show the sensor being touched. Notice the steep edges showing the response speed ($\sim 150\text{-}200\text{ms}$).

System Integration

The capacitive pad is the system's only user input: it starts and stops a logging session. When toggled on, the system begins sampling all three sensing channels (DS18B20, thermistor bridge, LPS33HW) every 20 seconds and buffering results in RAM; toggling it off ends the session. It doesn't interact with sensor values; it's a pure state-machine gate that determines whether a session is actively being recorded.

Limitations and Performance

The chosen RC combination produces a very short baseline ($\sim 2 \mu\text{s}$), which sits close to the timer's granularity noise floor; this is exactly why the touched/untouched separation reads as a large relative jump ($2 \rightarrow 6+ \mu\text{s}$) despite being a small absolute one. Breadboard proximity also inflates the effective C_{int} , raising the baseline; a PCB migration would lower it but require recalibrating the $6 \mu\text{s}$ threshold. It was noticed that when switching to the battery-powered approach, not connected through USB to a laptop, the average charge time decreased. This is likely due to the reduced capacitance of the chassis ground. If a PCB migration were to occur, switching to a $1\text{M}\Omega$ resistor would be useful to gain more time steps between touched and non-touched states.

5. Other Sensors Used

DS18B20 (Pre-packaged Sensor #1)

Raw physics

A digital, factory-calibrated bandgap temperature sensor communicating over the Dallas 1-Wire protocol, with 12-bit resolution (0.0625 °C steps).

Output Characteristics

Datasheet-rated accuracy is ± 0.5 °C over -10 to +85 °C. During the water-bath calibration (CP2), raw error ran up to 1.5875 °C across the full 4.4 to 43.9 °C test range, but only up to 0.2875 °C within the 20-30 °C operating band the system actually lives in. After applying the linear correction $T = 1.054 \cdot S - 1.296$, max error dropped to 0.134 °C and average percent error to 0.30%.

Expected vs. Actual

Actual pre-correction error was worse than the datasheet spec in absolute terms, but the corrected result comfortably beats it within the operating band, suggesting the raw nonlinearity, not fundamental sensor noise, was the dominant error source.

Guessed internal signal processing

Being a fully digital part, the DS18B20 almost certainly applies its own on-chip linearization/trim before reporting, consistent with how clean and linear ($R^2 = 0.999$) the raw output already looked before any processing was performed.

One limitation of the system

It's a single-point probe. Positioned away from the electronics inside the dough, it reports one location's temperature, not a true bulk average. Although the dough is well mixed at the start, the edges and bottom of the dough will cool faster than the center. This means that depending on the placement of the probe, it could measure many degrees off the dough's true average.

LPS33HW (Pre-packaged Sensor #2)

Raw physics

A MEMS piezoresistive absolute-pressure sensor with an integrated temperature channel, read digitally over I2C, rated to 1260 hPa.

Output Characteristics

Outdoor calibration against local weather-station data (with sea-level correction applied) showed the sensor reading consistently 0.02-0.09 inHg above the reported station value across two trials, with short-term noise of about ± 2 hPa in a still setting. A separate elevation test (measuring pressure drop across ~1020 ft of height) gave an average lapse rate within about 6% of the theoretical value.

Expected vs. Actual

The small, consistent positive offset (rather than random scatter) points to a minor systematic bias, not a broken sensor, plausibly from an uncorrected local weather-station discrepancy as much as the sensor itself. Individual elevation segments showed more scatter than

the full-height average, which makes sense given how small each segment's true pressure difference was relative to the sensor's own ± 2 hPa noise floor. Overall, the device appeared well calibrated.

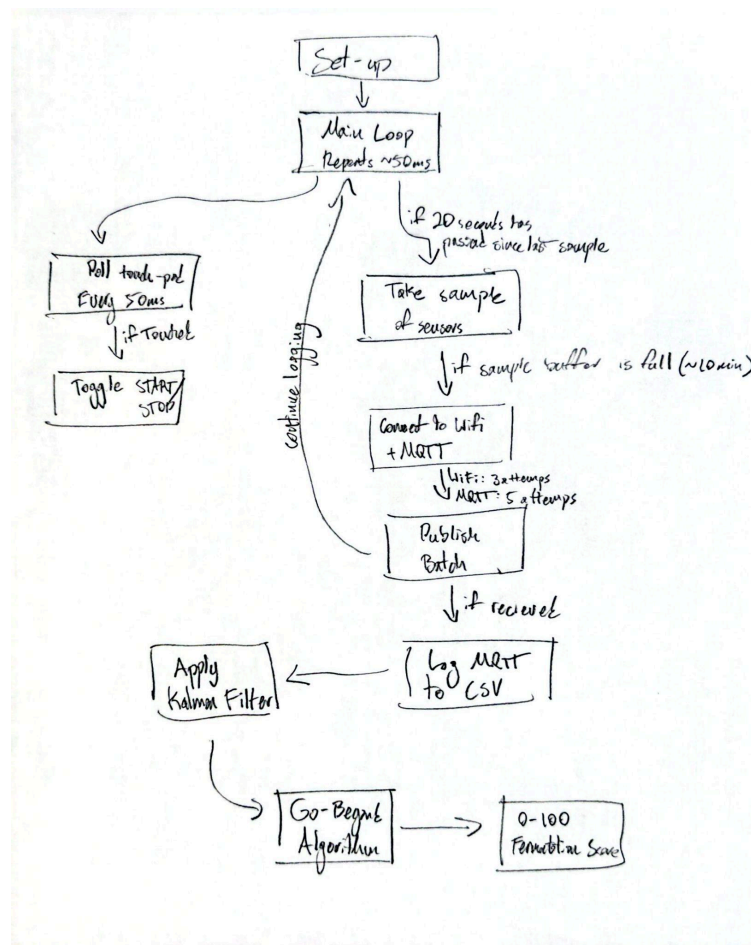
Guessed internal signal processing

The sensor most likely uses the mechanical bending of a silicon membrane to change the electrical resistance of piezoresistive elements. A current or voltage is run through or across these elements to determine the resistance change. This is likely digitally read, then adjusted with the onboard temperature sensor, and then sent over I2C.

One limitation of the system

One limitation of the system is that the temperature sensor never seemed accurate. The sensor seemed to read a couple of degrees below the raw value of the DS18B20 sensor. This means that the actual temperature of the LPS33HW was likely higher than was being read. This also means that the pressure it was reporting was also off because the system's digital calibration used that temperature measurement. However, it is possible that each sensor was calibrated to pressure values and not temperature, and therefore this has less of an effect.

6. Digital Logic and Algorithms



Algorithm Descriptions

Each sensor's protocol is initialized during set-up (One-Wire, I2C, GPIO pull-up). Then, during the sampling, each sensor's subfunction is called. A GPIO pin connected to the high voltage side of the thermistor bridge is activated, then 20 bridge readings over ~500 ms are taken and averaged into one reading. The DS18B20 and LPS33HW are each called in a process where 3 samples are taken and subsequently averaged. This exists because a single raw sample carries real jitter and random noise. A quick averaging process like this reduces the amount of random noise that is being recorded.

The Kalman filter algorithm applies a separate 1D Kalman filter to each data type (DS18B20 temperature, thermistor temperature, and LPS33HW pressure). The filter aids to further suppress the sample-to-sample jitter without introducing meaningful lag on a signal that only changes a few degrees over hours. This filter acts as another step to reduce random noise within the system.

The Go-Beyond algorithm starts by taking a temperature-adjusted pressure reading over time, zeroing it against the very first reading so what's left is pure pressure rise, and then fits that rise to a power-law model, $\Delta P(t) = a \cdot t^b$, using nonlinear least-squares regression. From that fitted curve, the algorithm projects the smooth model out to the actual end time of the run to get a de-noised total pressure rise, G , which becomes the raw measure of gas production. That raw

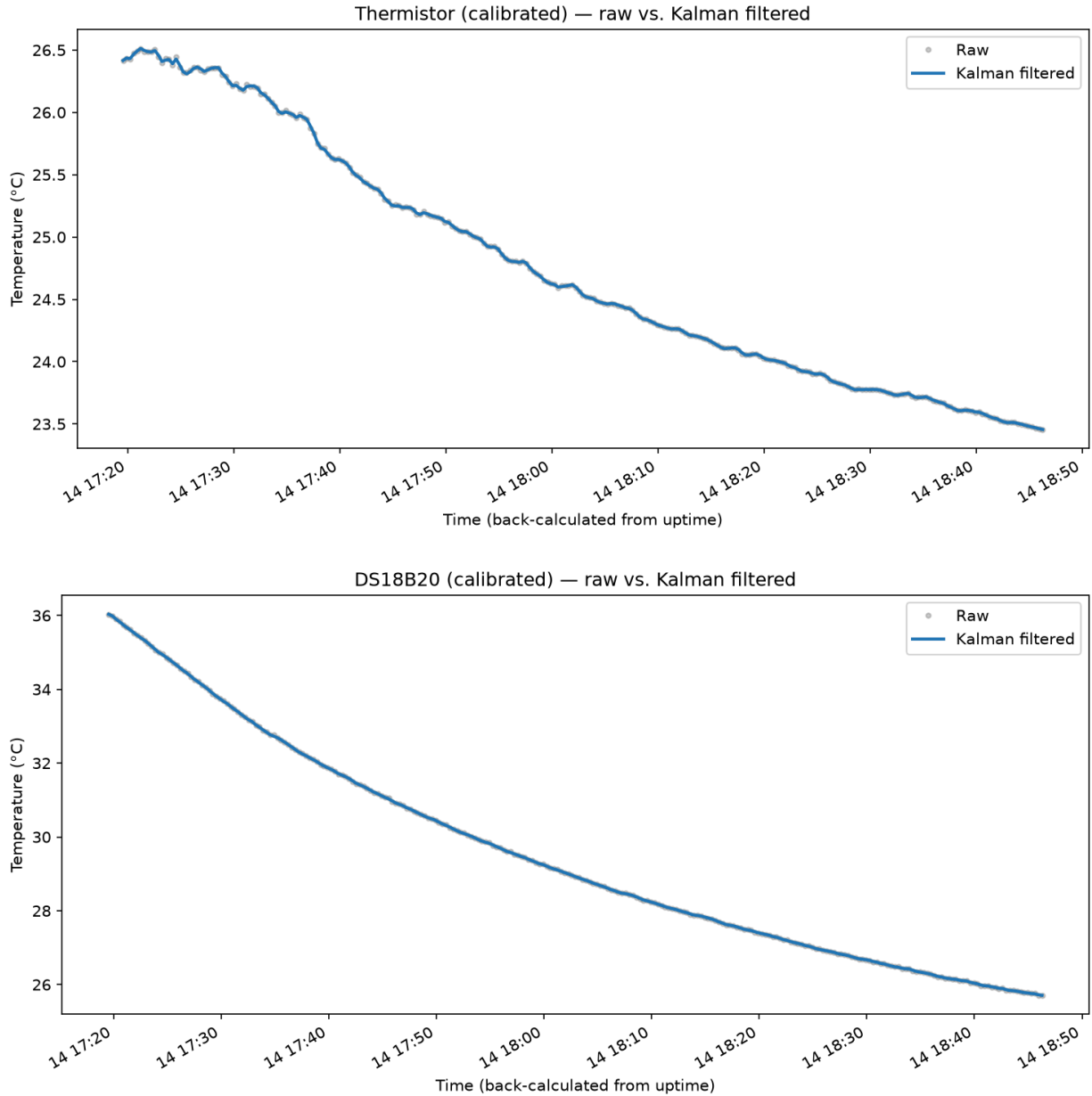
number is then normalized twice before it means anything comparable across bakes: first divided by grams of yeast used, so it reflects lift-per-gram rather than just total lift (which would unfairly reward simply using more yeast), and second divided by a Q10 temperature-fairness factor, which discounts runs that started warm (since warmth alone makes fermentation kinetics easier) and gives a bonus to runs that started cold but still performed well. Finally, a bounded shape multiplier derived from b nudges the score up slightly for an accelerating curve and down slightly for a decelerating one, since two runs with the same total gas output aren't equally healthy if one was still gaining momentum and the other was already stalling. The result is a single number, the FVI, that is then turned into a 0 to 100 score based on a good reference fermentation. This algorithm is helpful in determining the nature of a fermentation at a glance. It is not to be fully trusted as the algorithm isn't perfect and boils down a lot of data into one data point. However, it can be a good guide to determine which samples are worth looking at or which samples are consistent with each other.

Design Decision

The ESP32 was chosen for its combination of integrated WiFi, multiple ADC/I2C/GPIO channels, and deep-sleep support, all needed for the duty-cycled “sample every 20 s, publish every ~10 min” scheme that keeps battery draw low. An Arduino Uno would not have worked as well here: it has no native WiFi (requiring an added module and more power draw and complexity), a single-channel 10-bit ADC that would strain the signal chain, and no comparable low-power sleep mode to support the same duty cycle on a 350 mAh battery. It was mainly chosen because an ESP32 was on hand and fit the design well enough.

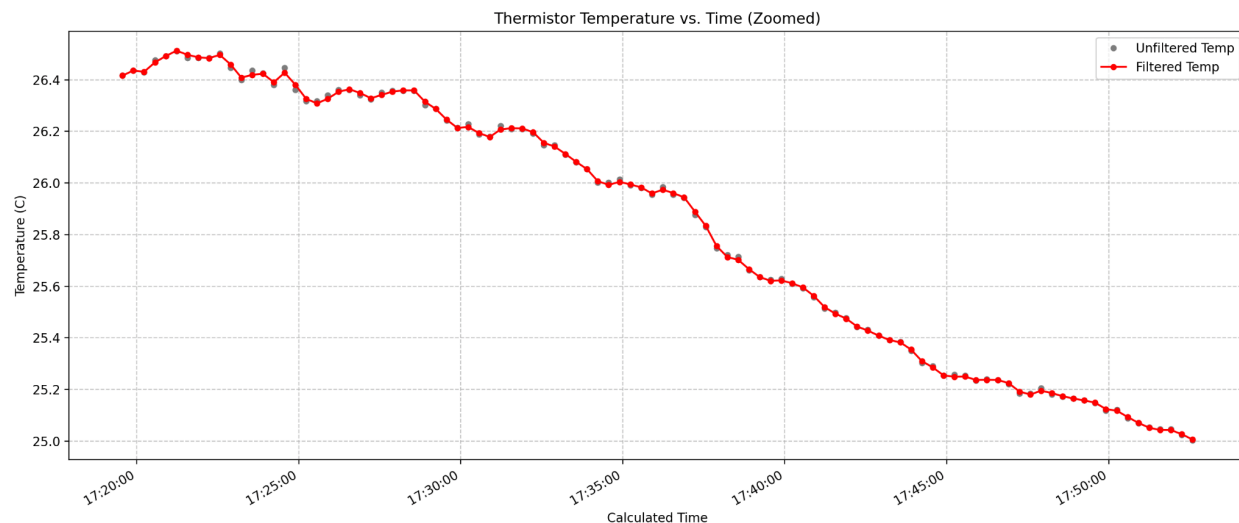
7. Final Design and Experimental Results

Result 1: Temperature Sensing Accuracy



Both channels tracked the same cooling trend in parallel and didn't fully settle until roughly three hours in, stabilizing near 22.8 °C, the DS18B20 held 22.75-22.81 °C over that window while the thermistor held 22.86-23.00 °C, a ~0.1-0.2 °C cross-channel offset that's within each sensor's own calibration error. The thermistor bridge shows visibly more sample-to-sample jitter than the DS18B20's smoother trace, consistent with the DS18B20 doing its own on-chip digitization versus the thermistor's fully analog front end. Against the <0.1 °C accuracy target from Section 1, the thermistor bridge clears it comfortably (0.0215 °C mean nonlinearity error), while the DS18B20's 0.134 °C max error sits just outside it.

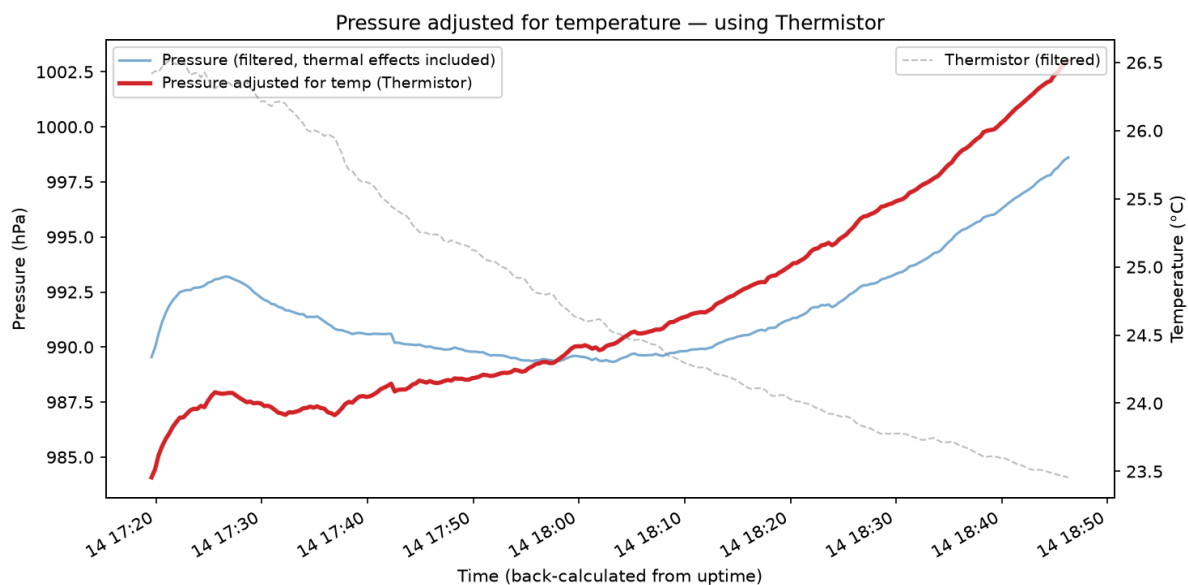
Result 2: Kalman Filter Impact on Real Fermentation Data

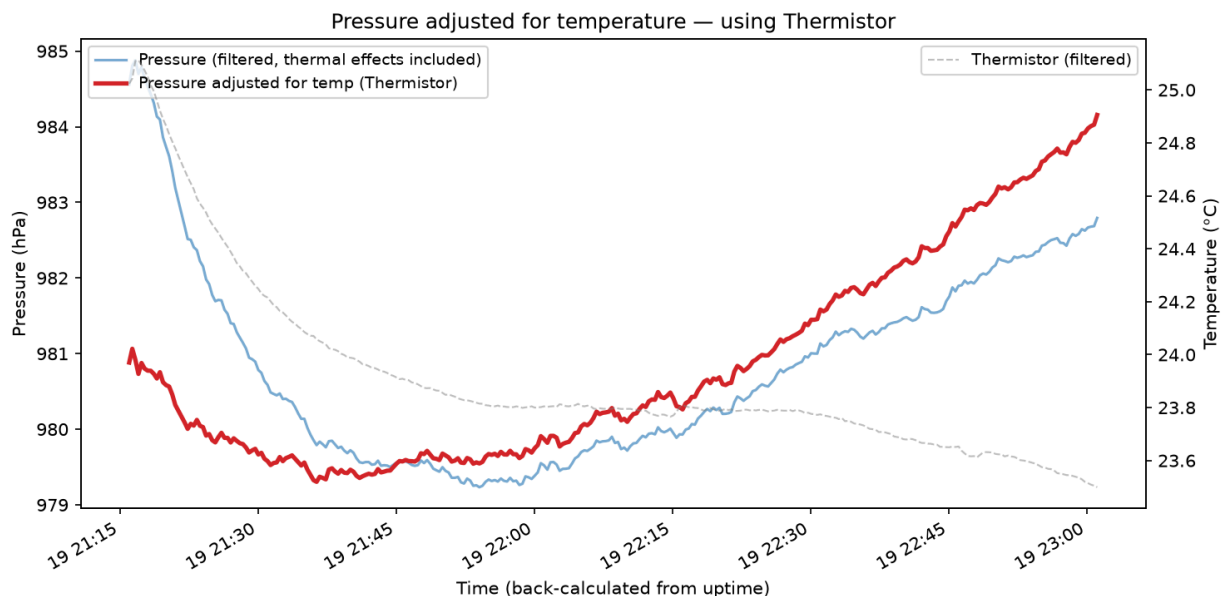


RMS sample-to-sample jump fell from 0.0205 °C (raw) to 0.0175 °C (filtered), a 14.7% reduction; residual variance from the local trend fell by 18.7%.

The filter produces a real but modest improvement on the finished system's actual 20-second sample rate: because consecutive raw readings are already close together at that cadence, there isn't much high-frequency noise left to remove, so the filter's main value is smoothing the occasional single-sample outlier rather than correcting large swings. This shows it's doing real, if modest, work rather than passing data through unchanged.

Result 3: Go-Beyond Algorithm





Above are two fermentation runs at different points on the score range: the reference run (top) shows adjusted pressure climbing from ~987 to over 1002 hPa in about 70 minutes with the slope visibly accelerating toward the end, and it scored 95/100, while the second run (bottom) shows a genuine ~25-minute lag phase before gas production picks up and then rises steadily but without acceleration, roughly a third of the reference's rate, earning a 61.7/100. This spread shows the Go-Beyond score isn't a simple pass/fail on whether pressure rose at all; it distinguishes strong, still-accelerating fermentation from a real but sluggish rise, which is exactly the "how far along, and how healthily" measure the project set out to build in Section 1.

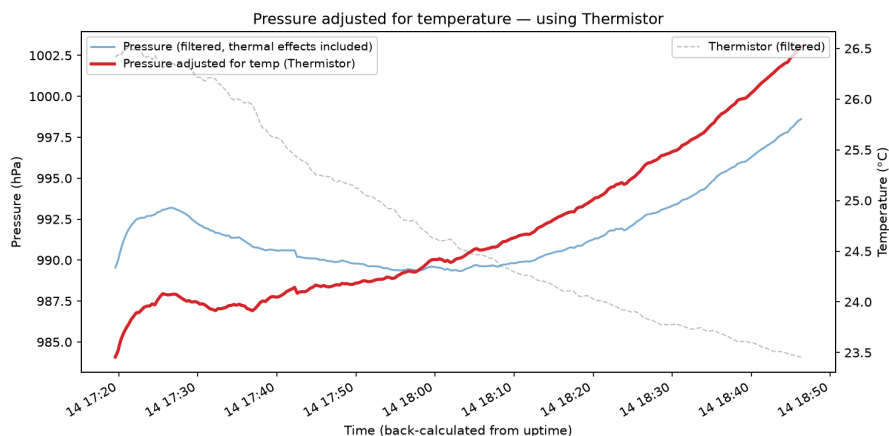
8. Failure Analysis

At the beginning of the project, the assumptions that getting a sealed pressure vessel wasn't that big of a concern. It should have been. I spent a couple of test trials with a plastic container to get a baseline idea of the system. I then moved on to a glass jar with a metal lid, but still had issues with imprecise pressure values.



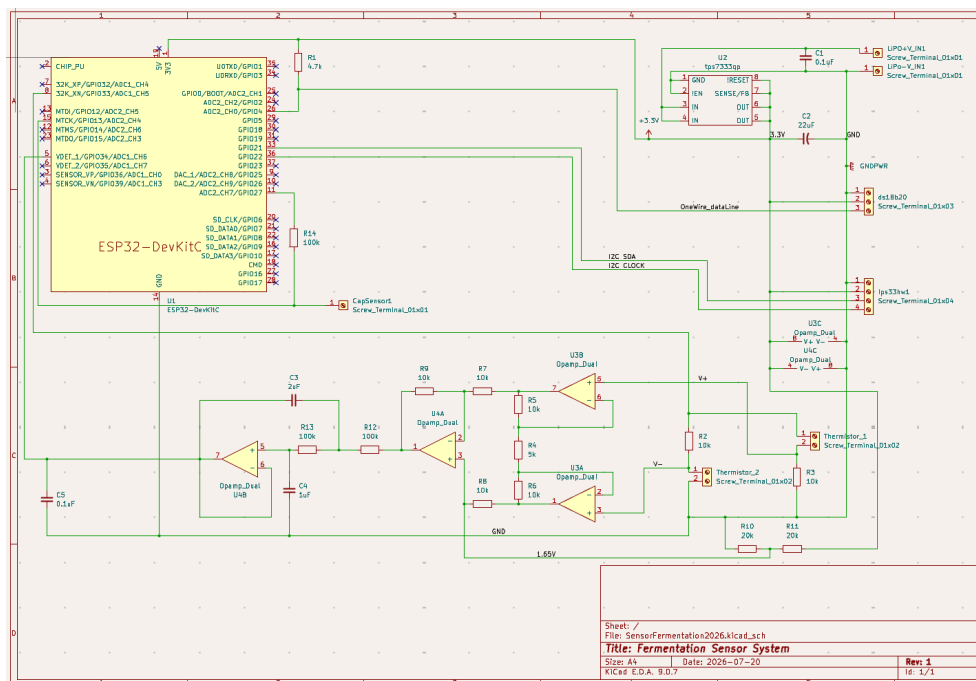
The dough had bubbles and looked like it was fermenting, so the seal had to be the problem. Originally, holes were drilled through the metal lid, through which the wires were fed, and the seal consisted of a silicone putty that was pressed onto the wires and into the gaps. Unfortunately, this wasn't enough. Subsequently, the putty on the inside of the lid was removed and replaced with hot glue, with scoring along the wires and metal to help adhesion.

This adaptation allowed the system to achieve reasonable readings. However, on the best run tested, there is still a slight pressure drop in the adjusted pressure, which could be cause for concern.

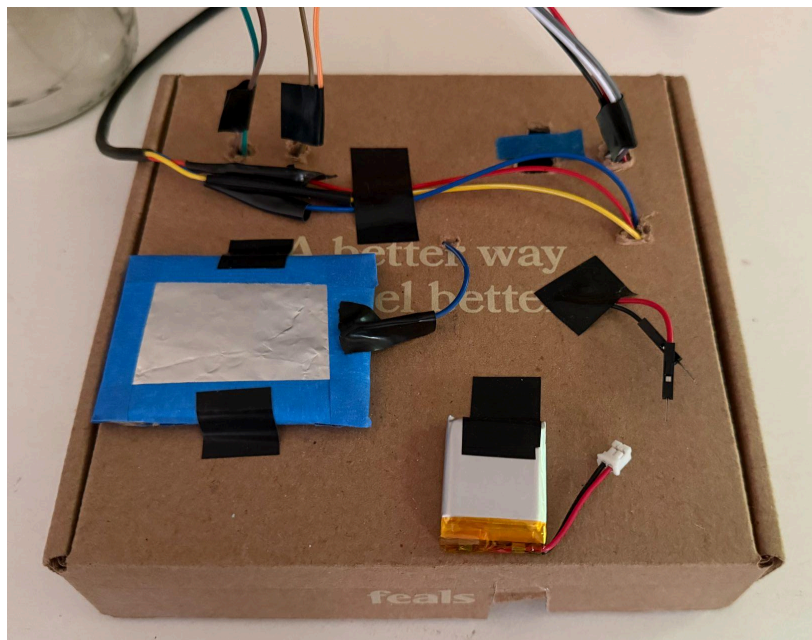


The slight drop after the initial temperature spike could be due to leakage in the system, suggesting it may not be entirely sealed. However, it may not be due to a leak. Both curves shown in 7.3 also show a brief dip in the adjusted trace during the fastest part of the initial temperature drop, before settling into a clean trend once the thermistor stabilizes. This could boil down to two causes. First, the thermistor's accuracy is meaningful off and not giving a true adjustment for the headspace pressure. Secondly, it is possible that the temperature gradient that exists within the jar is causing the thermistor to read significantly lower temperatures than the true average temperature. This would cause it to under-represent the true pressure in the vessel. In all likelihood, it is a combination of all three effects playing to misrepresent the pressure; however, meaningful data is still being collected, so the system is not a failure.

9. Physical Implementation



Off-board connections (sensors, battery, ESP32) use screw terminals rather than headers or JST connectors to tolerate stranded lead wire and repeated disconnect/reconnect during debugging without wearing out the board. The design uses wider 3.3V/GND traces for the ESP32's current demand, keeps components off the board edge for manufacturability, routes 3.3V and GND close together to reduce EMI, and places decoupling capacitors close to the LDO. It passed the DRC/ERC with zero violations.



Electronics live on breadboards in a small cardboard box, with the battery and touch pad mounted on top. The jar body is detachable for washing; the DS18B20 probe is positioned away from electronics so it can simply be wiped down.

Power Implementation

A 350 mAh, 3.7 V LiPo battery feeds a TPS7333QP LDO regulating down to 3.3 V. The LDO was chosen over a switching regulator specifically because the battery-to-rail voltage differential is small, and a switcher's noise would have degraded the sensitive thermistor-bridge measurement. The 20-second sample / ~10-minute publish duty cycle exists specifically to stretch this small battery's runtime.

A Physical Limitation

The breadboard-based construction itself. It kept the system serviceable while debugging, but it's the item that should be cut first. Trading a modest upfront PCB cost for a real gain in long-term reliability and repeatability. The breadboard has two big issues. The first is that it didn't hold wires that securely, so when trying to diagnose issues, it was necessary to be careful not to knock loose a wire. Second, the breadboard adds parasitic resistance and capacitance to the thermistor circuits, which is likely why it has some of the extra noise.

10. Course Reflection

Project Reflections

What assumption was most incorrect?

That achieving a reliable, fully sealed pressure vessel would be a straightforward mechanical detail rather than a persistent open problem; it remained the weakest link in the system all the way through CP5.

Where did theory deviate most from reality?

In the beginning, I posed this idea that the yeast would produce enough heat on its own to be noticeable. This was not the case, even in a controlled setting where the dough started at room temperature; I saw minor temperature changes, but they could have been due to changes in the ambient air temperature. I'm sure on larger scales this effect would have been more pronounced. With the slight temperature change because of the yeast, it would have been difficult to notice the temperature decrease of the dough cooling toward room temperature.

What design decision limited your performance?

Building the instrumentation amplifier from discrete op-amps (matched to within 10 Ω) rather than a dedicated in-amp IC, a decision forced by ordering parts before the Wheatstone-bridge approach was even settled on. A dedicated in-amp would likely have given better CMRR and stability. Although I am still pleased with how the thermistor channel turned out. It worked surprisingly well and had very few reliability issues. There were some runs where it jumped around more than the average noise, but this was not common.

What would you NOT do again?

If I did this again, the first thing I'd change is getting a PCB built before testing started. I kept holding off because I wasn't confident in the design yet and wanted room to change things, and being off-campus without my own soldering iron made revisions slower than they should've been. In hindsight, locking the design down earlier and accepting a PCB's constraints would have paid off in reliability.

I'm more torn on the Go-Beyond score itself; a single 0-100 number necessarily throws away information, and "good fermentation" depends on what the baker is even optimizing for. But I still think the underlying approach (a consistent dough portion, measured relative to itself) is the right one; the score is a first pass, not a finished answer.

Lecture Reflections

For the most part, I enjoyed the lectures. I think that the structure of this course is good and I thought that information was disseminated well. I will say that there is a LOT of information covered and a lot of things to learn.

11. References

Citations

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“IC OPAMP Dual Rail Power Connections - Schematic.” KiCad, KiCad, forum.kicad.info/t/ic-opamp-dual-rail-power-connections/41245. Accessed 28 July 2026.

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Shirur, Naveen. “Arduino Capacitive Sensor without Ground?.” Research Gate, Research Gate, 28 May 2019, www.researchgate.net/post/Arduino-capacitive-sensor-without-ground.

AI Use (Claude)

- Main use was coding
 - Fixing bugs
 - Adding comments
- Used to help brainstorm ideas at the beginning
- Along with the Texas Instruments Guide, it helped me choose an analog filter design
- Led me to discover sea-level pressure correction
- Helped to refine the Go-Beyond algorithm
 - Helped with the math used to fit the pressure curve
 - Helped refine the scoring system used
- I didn't really have any problems with AI use. I've learned that AI is best for code and general knowledge questions, and that's about it.

Appendix

Appendix A: Thermistor Datasheet

LINK: <https://www.vishay.com/docs/29154/ntcle213e3.pdf>

Appendix B: LDO Datasheet

LINK: <https://www.ti.com/lit/gpn/tps73>

Appendix C: Op-Amp Datasheet

LINK: <https://www.ti.com/lit/gpn/tlc2264>

Appendix D: LPS33HW Datasheet

LINK: <https://cdn-learn.adafruit.com/downloads/pdf/lps35hw-water-resistant-pressure-sensor.pdf>

Appendix E: 10k Ohm Resistor Datasheet

LINK: <https://www.vishay.com/docs/31018/cmfind.pdf>

Appendix F: DS18B20 Datasheet

LINK: <https://cdn.sparkfun.com/datasheets/Sensors/Temp/DS18B20.pdf>

Appendix G: ESP-32 Sensor Logging Code

```
#include <Wire.h>
#include <WiFi.h>
#include <PubSubClient.h>
#include <ArduinoJson.h>
#include <OneWire.h>
#include <DallasTemperature.h>

// ===== User configuration =====

const char* WIFI_SSID      = "WIFI_name";
const char* WIFI_PASSWORD = "WIFI_password";

const char* MQTT_BROKER    = "LAPTOP_IP";
const uint16_t MQTT_PORT   = 1883;
const char* MQTT_CLIENT_ID = "esp32-sensor-node";
const char* MQTT_TOPIC     = "sensors/node1/data";

// ===== Timing configuration =====

const unsigned long SAMPLE_INTERVAL_MS = 2000UL;           // one sample every 20
s

// Change from a 40-minute total duration to a 10-minute batch size
const unsigned long BATCH_DURATION_MS = 10UL * 60UL * 1000UL;

const size_t      MAX_SAMPLES          = BATCH_DURATION_MS / SAMPLE_INTERVAL_MS; // 30
samples

const unsigned long TOUCH_POLL_INTERVAL_MS = 50; // how often to check the touch pad

const unsigned long THERMISTOR_AVG_WINDOW_MS = 500; // keep thermistor powered
~0.5 s

const unsigned long THERMISTOR_SAMPLE_SPACING_MS = 20; // ~25 sub-samples per reading

const int N_DS18B20_SAMPLES = 3;
const int N_LPS33HW_SAMPLES = 3;

const unsigned long LPS33HW_SAMPLE_SPACING_MS = 45; // > 1/25Hz ODR period, for
independent samples
```

```

const size_t MQTT_BUFFER_SIZE = 12288; // big enough for a 60-sample JSON batch

// ===== Pin definitions =====
#define ONEWIRE_BUS_PIN    4
#define THERMISTOR_PIN    34    // ADC1 channel
#define THERMISTOR_PWR_PIN 33    // GPIO to power the thermistor circuit

#define TOUCH_SEND_PIN    27    // drives the RC charge
#define TOUCH_SENSE_PIN  13    // reads through resistor, connected to foil

// ===== Thermistor calibration =====
const float THERMISTOR_SLOPE_C_PER_MV = 0.00318766755178f;
const float THERMISTOR_OFFSET_C      = 19.8523791157f;

// ===== LPS33HW raw I2C driver =====
#define LPS33HW_ADDR    0x5D    // default addr (0x5C if SA0 pulled low)
#define WHO_AM_I_REG    0x0F
#define CTRL_REG1        0x10
#define PRESS_OUT_XL     (0x28 | 0x80) // OR 0x80 enables auto-increment
#define TEMP_OUT_L       (0x2B | 0x80)

bool initLPS33HW() {
    Wire.beginTransmission(LPS33HW_ADDR);
    Wire.write(WHO_AM_I_REG);
    Wire.endTransmission(false);
    Wire.requestFrom(LPS33HW_ADDR, 1);
    uint8_t whoAmI = Wire.read();

    if (whoAmI != 0xB1) {
        Serial.print("LPS33HW not found, WHO_AM_I = 0x");
        Serial.println(whoAmI, HEX);
        return false;
    }

    // ODR = 25 Hz, block data update enabled
    Wire.beginTransmission(LPS33HW_ADDR);

```

```

Wire.write(CTRL_REG1);
Wire.write(0x30);
Wire.endTransmission();
delay(10);
return true;
}

// Returns true on success, fills pressure_hPa and temp_C by reference
bool readPressureTemp(float &pressure_hPa, float &temp_C) {
    // --- Pressure (3 bytes, auto-increment) ---
    Wire.beginTransaction(LPS33HW_ADDR);
    Wire.write(PRESS_OUT_XL);
    if (Wire.endTransmission(false) != 0) return false;
    Wire.requestFrom(LPS33HW_ADDR, 3);
    if (Wire.available() < 3) return false;

    uint8_t x1 = Wire.read();
    uint8_t l  = Wire.read();
    uint8_t h  = Wire.read();

    int32_t pressureRaw = ((int32_t)h << 16) | ((int32_t)l << 8) | x1;
    if (pressureRaw & 0x800000) pressureRaw -= (1 << 24); // sign extend 24-bit
    pressure_hPa = pressureRaw / 4096.0f;

    // --- Temperature (2 bytes, auto-increment) ---
    Wire.beginTransaction(LPS33HW_ADDR);
    Wire.write(TEMP_OUT_L);
    if (Wire.endTransmission(false) != 0) return false;
    Wire.requestFrom(LPS33HW_ADDR, 2);
    if (Wire.available() < 2) return false;

    uint8_t t1 = Wire.read();
    uint8_t th = Wire.read();

    int16_t tempRaw = ((int16_t)th << 8) | t1;
    temp_C = tempRaw / 100.0f;

```

```

    return true;
}

// Average both pressure and temperature from the LPS33HW
void readAveragedLPS33HW(float &avgPressure, float &avgTemp) {
    float sumP = 0, sumT = 0;
    int validCount = 0;
    for (int i = 0; i < N_LPS33HW_SAMPLES; i++) {
        float p, t;
        if (readPressureTemp(p, t)) {
            sumP += p;
            sumT += t;
            validCount++;
        }
        if (i < N_LPS33HW_SAMPLES - 1) delay(LPS33HW_SAMPLE_SPACING_MS);
    }

    if (validCount > 0) {
        avgPressure = sumP / validCount;
        avgTemp = sumT / validCount;
    } else {
        avgPressure = NAN;
        avgTemp = NAN;
    }
}

// ===== DS18B20 =====
OneWire oneWire(ONEWIRE_BUS_PIN);
DallasTemperature ds18b20(&oneWire);

float readAveragedDS18B20() {
    float sum = 0;
    int validCount = 0;
    for (int i = 0; i < N_DS18B20_SAMPLES; i++) {
        ds18b20.requestTemperatures(); // blocking, ~750 ms at default 12-bit resolution
    }
}

```

```

float t = ds18b20.getTempCByIndex(0);

if (t != DEVICE_DISCONNECTED_C) {

    sum += t;

    validCount++;

}

}

return validCount > 0 ? sum / validCount : NAN;

}

// ===== Thermistor =====

float milliVoltsToTempC(float milliVolts) {

    return THERMISTOR_SLOPE_C_PER_MV * milliVolts + THERMISTOR_OFFSET_C;

}

// Power the thermistor divider for ~THERMISTOR_AVG_WINDOW_MS and average
// several ADC readings taken during that window, then power it back down.

float readAveragedThermistor(uint32_t &avgMilliVolts) {

    digitalWrite(THERMISTOR_PWR_PIN, HIGH);

    delay(10); // let the divider settle before the first reading

    uint32_t sumMv = 0;
    uint32_t count = 0;
    unsigned long start = millis();
    while (millis() - start < THERMISTOR_AVG_WINDOW_MS) {

        sumMv += analogReadMilliVolts(THERMISTOR_PIN);

        count++;

        delay(THERMISTOR_SAMPLE_SPACING_MS);

    }

    digitalWrite(THERMISTOR_PWR_PIN, LOW);

    avgMilliVolts = (count > 0) ? (sumMv / count) : 0;

    return milliVoltsToTempC((float)avgMilliVolts);

}

// ===== Capacitive touch (logging start/stop switch) =====

```

```

int baseline = 0;
int threshold = 0;
int elapsed_1 = 0;
int elapsed_2 = 0;
int elapsed_3 = 0;
int average_elapsed = 0;
bool pressed_last = false;
bool loggingEnabled = false;

// FIX: Debounce state for the touch switch. A single noisy reading above
// threshold can no longer register as a touch by itself - it has to stay
// above threshold for TOUCH_DEBOUNCE_POLLS consecutive polls in a row
// (tracked via an up/down counter), and has to fully drop back below
// threshold before another touch can be registered. This is what prevents
// one physical touch from being misread as touch-release-touch and
// toggling the logging state two or three times in a row.
const int TOUCH_DEBOUNCE_POLLS = 1; // ~200 ms of sustained signal at 50 ms/poll
int touchDebounceCount = 0;

int charge_discharge() {
    // Discharge
    pinMode(TOUCH_SENSE_PIN, OUTPUT);
    digitalWrite(TOUCH_SENSE_PIN, LOW);
    delayMicroseconds(10);
    pinMode(TOUCH_SENSE_PIN, INPUT);

    // Charge and time
    digitalWrite(TOUCH_SENSE_PIN, HIGH);
    unsigned long start = micros();
    while (digitalRead(TOUCH_SENSE_PIN) == LOW) {
        if (micros() - start > 5000) break; // timeout so we never hang
    }
    unsigned long elapsed = micros() - start;
    digitalWrite(TOUCH_SENSE_PIN, LOW);

    return elapsed;
}

```

```

}

void toggleLogging(); // defined near the sampling logic below

void pollTouchSensor() {
    elapsed_1 = charge_discharge();
    average_elapsed = (elapsed_1 + elapsed_2 + elapsed_3) / 3;
    // Serial.println(average_elapsed); // uncomment to tune threshold/margin below

    // FIX: up/down debounce counter instead of a single-sample decision.
    // Above threshold -> counter climbs (capped); below -> counter falls.
    // Only a fully-saturated counter counts as "touched", and it must fully
    // drain back to 0 before a new touch can be registered. This absorbs
    // the kind of +/-few-microsecond jitter that a single-sample threshold
    // (like the old "baseline + 3") would misread as press/release cycles.
    if (average_elapsed > threshold) {
        if (touchDebounceCount < TOUCH_DEBOUNCE_POLLS) touchDebounceCount++;
    } else {
        if (touchDebounceCount > 0) touchDebounceCount--;
    }

    bool touched = (touchDebounceCount >= TOUCH_DEBOUNCE_POLLS);

    if (touched && !pressed_last) {
        toggleLogging();
        pressed_last = true;
    } else if (touchDebounceCount == 0) {
        pressed_last = false;
    }

    elapsed_3 = elapsed_2;
    elapsed_2 = elapsed_1;
}

// ===== Sample buffer =====
struct SensorSample {

```

```

uint32_t t_ms;

float    pressure_hPa;
float    lps33hw_tempC;
float    ds18b20_tempC;
float    thermistor_tempC;
uint32_t thermistor_mV;
};

SensorSample sampleBuffer[MAX_SAMPLES];

size_t sampleCount = 0;

unsigned long loggingStartMillis = 0;
unsigned long lastSampleMillis = 0;

// ===== Connectivity =====
WiFiClient wifiClient;
PubSubClient mqttClient(wifiClient);

bool connectWiFi() {
    const unsigned long WIFI_CONNECT_TIMEOUT_MS = 15000;
    const int MAX_WIFI_ATTEMPTS = 3;

    for (int attempt = 1; attempt <= MAX_WIFI_ATTEMPTS; attempt++) {
        WiFi.mode(WIFI_STA);
        delay(300);
        WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
        Serial.printf("Connecting to WiFi (attempt %d/%d)", attempt, MAX_WIFI_ATTEMPTS);

        unsigned long start = millis();
        while (WiFi.status() != WL_CONNECTED && millis() - start < WIFI_CONNECT_TIMEOUT_MS)
        {
            delay(500);
            Serial.print(".");
        }

        if (WiFi.status() == WL_CONNECTED) {
            Serial.println("\nWiFi connected.");
        }
    }
}

```

```

        return true;
    }

    // NEW: print the actual status code so we know *why* it failed
    Serial.printf("\nWiFi connect attempt timed out. status=%d (", WiFi.status());
    switch (WiFi.status()) {
        case WL_NO_SSID_AVAIL: Serial.print("NO_SSID_AVAIL - AP not found/out of
range"); break;

        case WL_CONNECT_FAILED: Serial.print("CONNECT_FAILED - wrong password or auth
rejected"); break;

        case WL_CONNECTION_LOST: Serial.print("CONNECTION_LOST"); break;

        case WL_DISCONNECTED: Serial.print("DISCONNECTED - still negotiating/no
reply"); break;

        case WL_IDLE_STATUS: Serial.print("IDLE_STATUS - stuck before scan even
completed"); break;

        default: Serial.print("other"); break;
    }
    Serial.println(")");

    WiFi.disconnect(true);
    delay(500);
}

Serial.println("WiFi failed after all attempts.");
return false;
}

bool connectMQTT() {
    const int MAX_MQTT_ATTEMPTS = 5;
    mqttClient.setBufferSize(MQTT_BUFFER_SIZE);
    mqttClient.setServer(MQTT_BROKER, MQTT_PORT);

    for (int attempt = 1; attempt <= MAX_MQTT_ATTEMPTS; attempt++) {
        Serial.printf("Connecting to MQTT broker (attempt %d/%d)...", attempt,
MAX_MQTT_ATTEMPTS);
        if (mqttClient.connect(MQTT_CLIENT_ID)) {
            Serial.println("connected");
            return true;
        }
    }
}

```

```

    }

    Serial.printf("failed, rc=%d, retrying in 2s\n", mqttClient.state());
    delay(2000);
}

Serial.println("MQTT failed after all attempts.");
return false;
}

void disconnectNetwork() {
    mqttClient.disconnect();
    delay(50); // allow the MQTT disconnect packet to send
    WiFi.disconnect(true);
    WiFi.mode(WIFI_OFF);
    Serial.println("Network disconnected. Radio OFF.");
}

void sendBatch() {
    if (sampleCount == 0) {
        Serial.println("Nothing to send.");
        return;
    }

    if (!connectWiFi()) {
        Serial.println("Skipping this send - will try again next cycle.");
        disconnectNetwork(); // make sure radio is off, don't clear sampleCount
        return;
    }

    if (!connectMQTT()) {
        Serial.println("Skipping this send - will try again next cycle.");
        disconnectNetwork();
        return; // sampleCount NOT cleared - data preserved for retry
    }

    DynamicJsonDocument doc(MQTT_BUFFER_SIZE);
    doc["node_id"] = MQTT_CLIENT_ID;

```

```

doc["session_start_ms"] = loggingStartMillis;
doc["sample_count"]     = sampleCount;

JSONArray samples = doc.createNestedArray("samples");
for (size_t i = 0; i < sampleCount; i++) {
    JsonObject o = samples.createNestedObject();
    o["t_ms"]    = sampleBuffer[i].t_ms;
    o["p_hPa"]   = sampleBuffer[i].pressure_hPa;
    o["lps_C"]   = sampleBuffer[i].lps33hw_tempC;
    o["ds_C"]    = sampleBuffer[i].ds18b20_tempC;
    o["th_C"]    = sampleBuffer[i].thermistor_tempC;
    o["th_mV"]   = sampleBuffer[i].thermistor_mV;
}

static char payload[MQTT_BUFFER_SIZE];
size_t len = serializeJson(doc, payload, sizeof(payload));

if (mqttClient.publish(MQTT_TOPIC, payload, len)) {
    Serial.printf("Published batch of %u samples (%u bytes)\n", (unsigned)sampleCount,
(unsigned)len);
} else {
    Serial.println("MQTT publish failed (check broker connection & payload size).");
}

mqttClient.loop();
delay(100); // give the stack a moment to flush the outbound packet

disconnectNetwork();

sampleCount = 0; // clear the buffer once it's been sent
}

// ===== Sampling / logging session logic =====
void takeSample() {
    if (sampleCount >= MAX_SAMPLES) return; // safety net; shouldn't trigger, see fix
    below

```

```

SensorSample s;
s.t_ms = millis() - loggingStartMillis;

// Get both values from the LPS33HW
float p, t;
readAveragedLPS33HW(p, t);
s.pressure_hPa = p;
s.lps33hw_tempC = t;

s.ds18b20_tempC = readAveragedDS18B20();

uint32_t mv;
s.thermistor_tempC = readAveragedThermistor(mv);
s.thermistor_mV = mv;

sampleBuffer[sampleCount++] = s;

Serial.printf("Sample %u/%u P=%.4f hPa LPS_T=%.4f C DS18B20=%.4f C Therm=%.4f C
Therm=%u mV Clock=%u ms\n",
              (unsigned)sampleCount, (unsigned)MAX_SAMPLES,
              s.pressure_hPa, s.lps33hw_tempC, s.ds18b20_tempC, s.thermistor_tempC,
s.thermistor_mV, s.t_ms);

// FIX: this is the missing piece that implemented the documented
// "session runs the full 40 minutes -> auto-send and stop" behavior.
// Previously nothing checked for the buffer filling up, so once
// sampleCount hit MAX_SAMPLES the function above (the early "return;")
// just silently no-op'd forever - logging never stopped and the batch
// was never sent on its own; it required a manual touch to notice.
if (sampleCount >= MAX_SAMPLES) {
    Serial.println("10-minute mark reached - sending batch and continuing.");
    sendBatch(); // This sends the data and resets sampleCount back to 0

    // Notice that `loggingEnabled = false;` has been removed.
    // This allows the ESP32 to seamlessly start filling the next 10-minute batch!

```

```

}
}

void toggleLogging() {
    if (!loggingEnabled) {
        sampleCount = 0;
        loggingStartMillis = millis();
        lastSampleMillis = loggingStartMillis - SAMPLE_INTERVAL_MS; // sample right away
        loggingEnabled = true;
        Serial.println("Logging STARTED.");
    } else {
        Serial.println("Logging STOPPED (touch) - sending collected data.");
        loggingEnabled = false;
        sendBatch();
    }
}

// ===== Setup / Loop =====
unsigned long lastTouchPollMillis = 0;

void setup() {
    Serial.begin(115200);

    // 1. Wait 3 seconds so the Serial Monitor has time to catch the first messages
    delay(3000);
    Serial.println("\n\n--- ESP32 SYSTEM STARTUP ---");

    Serial.println("[1] Setting up pins...");
    pinMode(THERMISTOR_PWR_PIN, OUTPUT);
    digitalWrite(THERMISTOR_PWR_PIN, LOW);
    pinMode(TOUCH_SEND_PIN, OUTPUT);
    digitalWrite(TOUCH_SEND_PIN, LOW);

    Serial.println("[2] Initializing I2C bus...");
    Wire.begin();

    // Set a timeout so the ESP32 doesn't freeze forever if a sensor is stuck

```

```

Wire.setTimeout(150);

Serial.println("[3] Connecting to LPS33HW...");
while (!initLPS33HW()) {
    Serial.println("Retrying LPS33HW init... (Is it plugged in?)");
    delay(1000);
}

Serial.println("[4] Initializing DS18B20...");
ds18b20.begin();

Serial.println("[5] Configuring ADC...");
analogReadResolution(12);
analogSetPinAttenuation(THERMISTOR_PIN, ADC_11db);

Serial.println("[6] Calibrating touch pad. Keep hands away...");
delay(3000); // Give the power rails and your hands time to clear

long baseline_sum = 0;
for (int i = 0; i < 10; i++) {
    baseline_sum += charge_discharge();
    delay(10);
}

baseline = baseline_sum / 10;

// FIX: the old threshold ("baseline + 3" microseconds) was only a few
// microseconds of margin, which is well within the normal run-to-run
// jitter of an RC charge-time reading. That made the pad register
// phantom touches from ordinary electrical noise, which is what was
// causing logging to start/stop on its own. A touch adds real body
// capacitance, so it should shift the reading by a lot more than
// measurement noise does - a percentage-based margin is more robust
// than a fixed handful of microseconds, and scales automatically with
// your specific pad/resistor/RC circuit.
//

```

```

// Tune this on your hardware: uncomment the Serial.println(average_elapsed)
// line in pollTouchSensor(), watch the values with the pad untouched vs.
// touched, and set TOUCH_MARGIN_PERCENT comfortably between the two
// (roughly 1/3 to 1/2 of the way from baseline to a touched reading).
const int TOUCH_MARGIN_PERCENT = 20;    // touch trigger = baseline + 20%
const int TOUCH_MIN_MARGIN      = 1;    // floor, in case baseline is very small
int margin = (baseline * TOUCH_MARGIN_PERCENT) / 100;
if (margin < TOUCH_MIN_MARGIN) margin = TOUCH_MIN_MARGIN;
threshold = baseline + margin;

elapsed_2 = elapsed_3 = baseline;

Serial.print("Touch baseline: ");
Serial.print(baseline);
Serial.print(" | Threshold: ");
Serial.println(threshold);

WiFi.mode(WIFI_OFF);
Serial.println("--- SETUP COMPLETE. Radio OFF. Touch pad to start. ---");
// delay(1000);
// loggingEnabled = true;
}

void loop() {
    unsigned long now = millis();

    // Poll the touch pad frequently so start/stop stays responsive
    if (now - lastTouchPollMillis >= TOUCH_POLL_INTERVAL_MS) {
        lastTouchPollMillis = now;
        pollTouchSensor();
    }

    // While a session is running, take one averaged sample every SAMPLE_INTERVAL_MS
    if (loggingEnabled && (now - lastSampleMillis >= SAMPLE_INTERVAL_MS)) {
        lastSampleMillis = now;
        takeSample();
    }
}

```

```
}  
delay(50);  
}
```

Appendix H: MQTT Logging Code

```
import csv
import json
import os
import signal
import sys
from datetime import datetime

import paho.mqtt.client as mqtt

# ===== Configuration =====
MQTT_BROKER = "localhost"          # change if the broker isn't on this machine
MQTT_PORT = 1883
MQTT_TOPIC = "sensors/node1/data"

CSV_FILE = "sensor_log.csv"
CSV_FIELDNAMES = [
    "timestamp",
    "pressure_hPa",
    "lps33hw_tempC",
    "ds18b20_tempC",
    "thermistor_mV",
    "thermistor_tempC",
    "uptime_ms",
]

def ensure_csv_header(path):
    """Create the CSV file with a header row if it doesn't already exist."""
    file_exists = os.path.isfile(path)
    if not file_exists:
        with open(path, mode="w", newline="") as f:
            writer = csv.DictWriter(f, fieldnames=CSV_FIELDNAMES)
            writer.writeheader()
```

```

def append_row(path, row):
    """Append a single reading to the CSV file."""
    with open(path, mode="a", newline="") as f:
        writer = csv.DictWriter(f, fieldnames=CSV_FIELDNAMES)
        writer.writerow(row)

def on_connect(client, userdata, flags, reason_code, properties=None):
    if reason_code == 0:
        print(f"Connected to MQTT broker at {MQTT_BROKER}:{MQTT_PORT}")
        client.subscribe(MQTT_TOPIC)
        print(f"Subscribed to topic: {MQTT_TOPIC}")
    else:
        print(f"Failed to connect, reason code: {reason_code}")

def on_message(client, userdata, msg):
    try:
        payload = json.loads(msg.payload.decode("utf-8"))
    except (UnicodeDecodeError, json.JSONDecodeError) as e:
        print(f"Skipping malformed message: {e}")
        return

    # The ESP32 sends an array of readings under the "samples" key
    samples = payload.get("samples", [])

    if not samples:
        print("Received payload, but no samples were found.")
        return

    # Loop through the batch and write each sample to the CSV
    for sample in samples:
        row = {
            "timestamp": datetime.now().isoformat(timespec="seconds"),
            "pressure_hPa": sample.get("p_hPa", ""),

```

```

        "lps33hw_tempC": sample.get("lps_C", ""), # <-- ADD THIS LINE
        "ds18b20_tempC": sample.get("ds_C", ""),
        "thermistor_mV": sample.get("th_mV", ""),
        "thermistor_tempC": sample.get("th_C", ""),
        "uptime_ms": sample.get("t_ms", ""),
    }

    append_row(CSV_FILE, row)
    print(f"Logged: {row}")

def main():
    ensure_csv_header(CSV_FILE)

    client = mqtt.Client(callback_api_version=mqtt.CallbackAPIVersion.VERSION2)
    client.on_connect = on_connect
    client.on_message = on_message

    client.connect(MQTT_BROKER, MQTT_PORT, keepalive=60)

    def handle_sigint(sig, frame):
        print("\nStopping logger, disconnecting...")
        client.disconnect()
        sys.exit(0)

    signal.signal(signal.SIGINT, handle_sigint)

    print(f"Logging to {os.path.abspath(CSV_FILE)} - press Ctrl+C to stop.")
    client.loop_forever()

if __name__ == "__main__":
    main()

```

Appendix I: Kalman Filter Code

```

import argparse
import os
import sys

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

REQUIRED_COLUMNS = [
    "timestamp", "pressure_hPa", "lps33hw_tempC", "ds18b20_tempC",
    "thermistor_mV", "thermistor_tempC", "uptime_ms",
]

# ----- #
# Kalman filter: constant-velocity ("trend") model
# ----- #

class ConstantVelocityKalman:
    """
    1D Kalman filter with state = [value, rate_of_change].

    F(dt) = [[1, dt], [0, 1]]
    Q(dt) = q * [[dt^4/4, dt^3/2], [dt^3/2, dt^2]] (discretized white-noise
        acceleration model -- standard choice for a "trend" filter with
        irregular time steps)
    H = [1, 0]
    """

    def __init__(self, initial_value, measurement_variance, process_variance,
                 initial_rate=0.0, initial_value_variance=None,
                 initial_rate_variance=1.0):
        self.R = measurement_variance
        self.q = process_variance
        if initial_value_variance is None:

```

```

        initial_value_variance = measurement_variance * 10

        self.x = np.array([[initial_value], [initial_rate]], dtype=float)
        self.P = np.diag([initial_value_variance, initial_rate_variance]).astype(float)
        self.H = np.array([[1.0, 0.0]])

    def predict(self, dt):
        dt = max(dt, 1e-6)

        F = np.array([[1.0, dt], [0.0, 1.0]])

        q = self.q

        Q = q * np.array([
            [dt**4 / 4, dt**3 / 2],
            [dt**3 / 2, dt**2],
        ])

        self.x = F @ self.x
        self.P = F @ self.P @ F.T + Q

    def update(self, z):
        H = self.H

        y = z - (H @ self.x)[0, 0]

        S = (H @ self.P @ H.T)[0, 0] + self.R

        K = (self.P @ H.T) / S

        self.x = self.x + K * y

        self.P = (np.eye(2) - K @ H) @ self.P

    @property
    def value(self):
        return self.x[0, 0]

    @property
    def rate(self):
        return self.x[1, 0]

def kalman_filter_series(times_sec, values, measurement_variance, process_variance):
    """Run the constant-velocity Kalman filter over a time-ordered series."""
    values = np.asarray(values, dtype=float)

```

```

times_sec = np.asarray(times_sec, dtype=float)
filtered = np.zeros_like(values)

kf = ConstantVelocityKalman(values[0], measurement_variance, process_variance)
filtered[0] = kf.value
for i in range(1, len(values)):
    dt = times_sec[i] - times_sec[i - 1]
    kf.predict(dt)
    kf.update(values[i])
    filtered[i] = kf.value
return filtered

def temperature_adjusted_pressure(pressure_filtered_hPa, temp_filtered_C, ref_temp_C):
    """
    Rescale pressure to a reference temperature using Gay-Lussac's Law
    (P / T_kelvin = constant at fixed volume/moles):

    
$$P_{\text{adjusted}} = P_{\text{filtered}} * (T_{\text{ref\_K}} / T_{\text{filtered\_K}})$$


    This removes the pressure swing explained by thermal expansion alone,
    leaving a residual that's a better proxy for actual gas production.
    """
    T_K = temp_filtered_C + 273.15
    T_ref_K = ref_temp_C + 273.15
    return pressure_filtered_hPa * (T_ref_K / T_K)

def estimate_measurement_noise(df, column, batch_col="timestamp", fallback=None):
    """
    Estimate measurement variance from the spread of readings within each
    upload batch (rows sharing the same upload timestamp). Falls back to the
    overall series variance if no batch has more than one reading.
    """
    batch_variances = df.groupby(batch_col)[column].var(ddof=1).dropna()
    if len(batch_variances) > 0:

```

```

        return float(batch_variances.mean())

    if fallback is not None:
        return fallback

    overall_var = df[column].var(ddof=1)
    return float(overall_var) if pd.notna(overall_var) and overall_var > 0 else 1.0

# ----- #
# Data loading / time back-calculation / calibration
# ----- #
def load_and_prepare(csv_path):
    df = pd.read_csv(csv_path)

    missing = [c for c in REQUIRED_COLUMNS if c not in df.columns]
    if missing:
        raise ValueError(f"Input CSV is missing required column(s): {missing}")

    before = len(df)
    df = df.dropna(subset=["timestamp", "pressure_hPa", "ds18b20_tempC",
                          "thermistor_mV", "uptime_ms"]).copy()

    if len(df) < before:
        print(f"Warning: dropped {before - len(df)} row(s) with missing required values.")

    df["timestamp"] = pd.to_datetime(df["timestamp"])

    # --- back-calculate true measurement time ---
    # timestamp = upload time = time of the LAST reading in its batch
    df["batch_max_uptime_ms"] = df.groupby("timestamp")["uptime_ms"].transform("max")
    df["offset_ms"] = df["batch_max_uptime_ms"] - df["uptime_ms"]
    df["true_time"] = df["timestamp"] - pd.to_timedelta(df["offset_ms"], unit="ms")

    df = df.sort_values("true_time").reset_index(drop=True)
    df["t_sec"] = (df["true_time"] - df["true_time"].iloc[0]).dt.total_seconds()

    # --- calibration ---

```

```

df["ds18b20_tempC_corrected"] = 1.054 * df["ds18b20_tempC"] - 1.296
df["thermistor_tempC_calc"] = 0.003293 * df["thermistor_mV"] + 19.557

return df

# ----- #
# Plotting
# ----- #
def plot_channel(df, raw_col, filtered_col, ylabel, title, out_path):
    fig, ax = plt.subplots(figsize=(10, 5))
    ax.plot(df["true_time"], df[raw_col], "o", ms=3, alpha=0.4,
            color="tab:gray", label="Raw")
    ax.plot(df["true_time"], df[filtered_col], "-", lw=2,
            color="tab:blue", label="Kalman filtered")
    ax.set_xlabel("Time (back-calculated from uptime)")
    ax.set_ylabel(ylabel)
    ax.set_title(title)
    ax.legend()
    fig.autofmt_xdate()
    fig.tight_layout()
    fig.savefig(out_path, dpi=150)
    return fig

def plot_pressure_adjustment(df, temp_col, adjusted_col, temp_label, out_path):
    fig, ax1 = plt.subplots(figsize=(10, 5))

    ax1.plot(df["true_time"], df["pressure_hPa_filtered"], "-", lw=1.5,
            color="tab:blue", alpha=0.6, label="Pressure (filtered, thermal effects
included)")
    ax1.plot(df["true_time"], df[adjusted_col], "-", lw=2.5,
            color="tab:red", label=f"Pressure adjusted for temp ({temp_label})")
    ax1.set_xlabel("Time (back-calculated from uptime)")
    ax1.set_ylabel("Pressure (hPa)")
    ax1.legend(loc="upper left", fontsize=9)

```

```

ax2 = ax1.twinx()
ax2.plot(df["true_time"], df[temp_col], "--", lw=1, color="gray",
        alpha=0.5, label=f"{temp_label} (filtered)")
ax2.set_ylabel("Temperature (°C)")
ax2.legend(loc="upper right", fontsize=9)

ax1.set_title(f"Pressure adjusted for temperature – using {temp_label}")
fig.autofmt_xdate()
fig.tight_layout()
fig.savefig(out_path, dpi=150)
return fig

def plot_combined(df, out_path):
    fig, ax = plt.subplots(figsize=(11, 6))

    ax.plot(df["true_time"], df["fermentation_activity_index"], "-",
            lw=2.5, color="tab:red", label="Fermentation activity index")
    ax.plot(df["true_time"], df["temp_norm"], "--", lw=1, alpha=0.6,
            color="tab:orange", label="Normalized temp (ds18b20 + thermistor avg)")
    ax.plot(df["true_time"], df["pressure_norm"], "--", lw=1, alpha=0.6,
            color="tab:green", label="Normalized pressure")

    ax.axhline(0, color="black", lw=0.7, alpha=0.5)
    ax.set_xlabel("Time (back-calculated from uptime)")
    ax.set_ylabel("Z-score / index (unitless)")
    ax.set_title("Combined fermentation activity index (temp + pressure)")
    ax.legend()
    fig.autofmt_xdate()
    fig.tight_layout()
    fig.savefig(out_path, dpi=150)
    return fig

# ----- #

```

```

# Main

# ----- #

def main():

    parser = argparse.ArgumentParser(
        description="Kalman-filter fermentation sensor logs and plot a combined
activity index.")

    parser.add_argument("input_csv", help="Path to input CSV file")

    parser.add_argument("--output-dir", default="fermentation_output",
                        help="Directory to save filtered CSV + plots (default:
./fermentation_output)")

    parser.add_argument("--pressure-r", type=float, default=None,
                        help="Override measurement variance for pressure_hPa (default:
auto-estimated)")

    parser.add_argument("--pressure-q", type=float, default=1e-8,
                        help="Process noise intensity for pressure trend (default:
1e-8)")

    parser.add_argument("--ds18b20-r", type=float, default=None,
                        help="Override measurement variance for calibrated ds18b20
temp (default: auto-estimated)")

    parser.add_argument("--ds18b20-q", type=float, default=1e-7,
                        help="Process noise intensity for ds18b20 trend (default:
1e-7)")

    parser.add_argument("--thermistor-r", type=float, default=None,
                        help="Override measurement variance for calibrated thermistor
temp (default: auto-estimated)")

    parser.add_argument("--thermistor-q", type=float, default=1e-7,
                        help="Process noise intensity for thermistor trend (default:
1e-7)")

    parser.add_argument("--temp-weight", type=float, default=0.5,
                        help="Weight of the (averaged) temperature signal in the
combined index (default: 0.5)")

    parser.add_argument("--pressure-weight", type=float, default=0.5,
                        help="Weight of the pressure signal in the combined index
(default: 0.5)")

```

```

    parser.add_argument("--pressure-ref", choices=["mean", "first", "last"],
                        default="mean",
                        help="Reference point used for temperature-adjusted pressure:
"
                        "'mean'/'first'/'last' reading of that temperature
channel (default: mean)")

    parser.add_argument("--show", action="store_true",
                        help="Also display plots interactively (in addition to saving
them)")

    args = parser.parse_args()

    df = load_and_prepare(args.input_csv)
    os.makedirs(args.output_dir, exist_ok=True)

    # --- noise estimation ---
    pressure_r = args.pressure_r if args.pressure_r is not None else \
        estimate_measurement_noise(df, "pressure_hPa")
    ds18b20_r = args.ds18b20_r if args.ds18b20_r is not None else \
        estimate_measurement_noise(df, "ds18b20_tempC_corrected")
    thermistor_r = args.thermistor_r if args.thermistor_r is not None else \
        estimate_measurement_noise(df, "thermistor_tempC_calc")

    print("Measurement noise variance (R) in use:")
    print(f"  pressure_hPa          : {pressure_r:.6g}")
    print(f"  ds18b20_tempC_corrected   : {ds18b20_r:.6g}")
    print(f"  thermistor_tempC_calc     : {thermistor_r:.6g}")

    # --- Kalman filter each channel individually ---
    df["pressure_hPa_filtered"] = kalman_filter_series(
        df["t_sec"], df["pressure_hPa"], pressure_r, args.pressure_q)
    df["ds18b20_tempC_filtered"] = kalman_filter_series(
        df["t_sec"], df["ds18b20_tempC_corrected"], ds18b20_r, args.ds18b20_q)
    df["thermistor_tempC_filtered"] = kalman_filter_series(
        df["t_sec"], df["thermistor_tempC_calc"], thermistor_r, args.thermistor_q)

```

```

# --- combined fermentation activity index (ds18b20 + thermistor only; lps33hw temp
excluded) ---

df["avg_temp_filtered"] = df[["ds18b20_tempC_filtered",
"thermistor_tempC_filtered"]].mean(axis=1)

temp_std = df["avg_temp_filtered"].std(ddof=0)
pressure_std = df["pressure_hPa_filtered"].std(ddof=0)

df["temp_norm"] = (df["avg_temp_filtered"] - df["avg_temp_filtered"].mean()) /
(temp_std if temp_std > 0 else 1.0)

df["pressure_norm"] = (df["pressure_hPa_filtered"] -
df["pressure_hPa_filtered"].mean()) / (pressure_std if pressure_std > 0 else 1.0)

df["fermentation_activity_index"] = (
    args.temp_weight * df["temp_norm"] + args.pressure_weight * df["pressure_norm"]
)

# --- temperature-adjusted pressure (removes thermal-expansion component) ---
def pick_ref(series):
    if args.pressure_ref == "mean":
        return series.mean()
    elif args.pressure_ref == "first":
        return series.iloc[0]
    else:
        return series.iloc[-1]

ref_thermistor = pick_ref(df["thermistor_tempC_filtered"])
ref_ds18b20 = pick_ref(df["ds18b20_tempC_filtered"])

df["pressure_adjusted_thermistor"] = temperature_adjusted_pressure(
    df["pressure_hPa_filtered"], df["thermistor_tempC_filtered"], ref_thermistor)
df["pressure_adjusted_ds18b20"] = temperature_adjusted_pressure(
    df["pressure_hPa_filtered"], df["ds18b20_tempC_filtered"], ref_ds18b20)

print(f"\nTemperature-adjusted pressure reference ({args.pressure_ref}):")
print(f"  thermistor : {ref_thermistor:.4f} °C")
print(f"  ds18b20      : {ref_ds18b20:.4f} °C")

```

```

# --- save filtered CSV ---
out_cols = [
    "true_time", "timestamp", "uptime_ms",
    "pressure_hPa", "pressure_hPa_filtered",
    "ds18b20_tempC", "ds18b20_tempC_corrected", "ds18b20_tempC_filtered",
    "thermistor_mV", "thermistor_tempC_calc", "thermistor_tempC_filtered",
    "lps33hw_tempC", # reference only, unfiltered, not used in the index
    "avg_temp_filtered", "temp_norm", "pressure_norm",
    "fermentation_activity_index",
    "pressure_adjusted_thermistor", "pressure_adjusted_ds18b20",
]

out_csv_path = os.path.join(args.output_dir, "filtered_output.csv")
df[out_cols].to_csv(out_csv_path, index=False)
print(f"\nSaved filtered data: {out_csv_path}")

# --- individual plots ---
fig1 = plot_channel(df, "pressure_hPa", "pressure_hPa_filtered",
                    "Pressure (hPa)", "Pressure - raw vs. Kalman filtered",
                    os.path.join(args.output_dir, "pressure_filtered.png"))
fig2 = plot_channel(df, "ds18b20_tempC_corrected", "ds18b20_tempC_filtered",
                    "Temperature (°C)", "DS18B20 (calibrated) - raw vs. Kalman
filtered",
                    os.path.join(args.output_dir, "ds18b20_filtered.png"))
fig3 = plot_channel(df, "thermistor_tempC_calc", "thermistor_tempC_filtered",
                    "Temperature (°C)", "Thermistor (calibrated) - raw vs. Kalman
filtered",
                    os.path.join(args.output_dir, "thermistor_filtered.png"))

fig4 = plot_combined(df, os.path.join(args.output_dir,
"fermentation_activity_index.png"))

fig5 = plot_pressure_adjustment(df, "thermistor_tempC_filtered",
"pressure_adjusted_thermistor",
                    "Thermistor", os.path.join(args.output_dir,
"pressure_adjusted_thermistor.png"))

fig6 = plot_pressure_adjustment(df, "ds18b20_tempC_filtered",
"pressure_adjusted_ds18b20",
                    "DS18B20", os.path.join(args.output_dir,
"pressure_adjusted_ds18b20.png"))

print("Saved plots:")

```

```
for name in ["pressure_filtered.png", "ds18b20_filtered.png",
            "thermistor_filtered.png", "fermentation_activity_index.png",
            "pressure_adjusted_thermistor.png", "pressure_adjusted_ds18b20.png"]:
    print(f" {os.path.join(args.output_dir, name)}")

if args.show:
    plt.show()
else:
    plt.close("all")

if __name__ == "__main__":
    main()
```

Call with:

python file_name.py input.csv

Appendix J: Go-Beyond Algorithm

```
import numpy as np
import pandas as pd
from scipy.optimize import curve_fit

def power_law(t, a, b):
    return a * np.power(t, b)

def fit_pressure_curve(csv_path, pressure_col="pressure_adjusted_thermistor",
                       time_col="true_time"):
    """Fit  $\Delta P(t) = a * t^b$  to one fermentation run."""
    df = pd.read_csv(csv_path, parse_dates=[time_col])
    t0 = df[time_col].iloc[0]
    t_min = (df[time_col] - t0).dt.total_seconds() / 60.0

    p0 = df[pressure_col].iloc[0]
    y = (df[pressure_col] - p0).values
    t = t_min.values + 0.05 # avoid t=0 in the power law

    y_clipped = np.clip(y, 1e-6, None) # power law needs y > 0
    (a, b), _ = curve_fit(power_law, t, y_clipped, p0=[0.5, 1.0], maxfev=20000)

    pred = power_law(t, a, b)
    r2 = 1 - np.sum((y - pred) ** 2) / np.sum((y - y.mean()) ** 2)

    return {
        "a": a,
        "b": b,
        "r2": r2,
        "p0": p0,
        "t_end_min": t_min.iloc[-1],
        "start_dough_tempC": df["ds18b20_tempC_filtered"].iloc[0],
        "start_air_tempC": df["thermistor_tempC_filtered"].iloc[0],
    }
```

```

}

def fermentation_vigor_index(fit, yeast_g, sugar_g=None,
                             T_ref=25.0, Q10=2.0):
    """
    Turn a fitted curve into the Fermentation Vigor Index.

    fit      : dict returned by fit_pressure_curve()
    yeast_g   : grams of yeast used in this bake (REQUIRED)
    sugar_g   : grams of sugar used (optional, reported as a secondary
                "substrate efficiency" number, not folded into FVI)
    T_ref     : reference starting dough temperature in C for the Q10
                correction. Pick whatever you consider a "neutral"
                starting temp for your kitchen/proofer.
    Q10       : how much yeast activity roughly multiplies per 10C.
                2.0 is a standard textbook approximation for the
                15-35C range; tune it if you have calibration bakes.
    """
    a, b, t_end = fit["a"], fit["b"], fit["t_end_min"]
    T0 = fit["start_dough_tempC"]

    G = a * (t_end ** b) # smooth (fitted) total pressure rise, hPa

    temp_factor = Q10 ** ((T0 - T_ref) / 10.0)
    shape_multiplier = 1 + 0.5 * np.tanh(b - 1)

    vigor_per_gram_yeast = G / yeast_g
    temp_normalized_vigor = vigor_per_gram_yeast / temp_factor
    fvi = temp_normalized_vigor * shape_multiplier

    result = {
        "total_pressure_rise_hPa": G,
        "kinetics_exponent_b": b,
        "shape_multiplier": shape_multiplier,
        "temp_factor": temp_factor,
    }

```

```

        "vigor_per_gram_yeast": vigor_per_gram_yeast,
        "FVI": fvi,
    }
    if sugar_g:
        result["vigor_per_gram_sugar"] = G / sugar_g
    return result

# --- Teaspoon -> gram conversions, since baking measurements are usually
# volumetric. These are standard culinary approximations (+/- ~10% depending
# on how packed the spoon is) -- swap in a kitchen scale reading if you have
# one, it'll be more precise than these.
SUGAR_G_PER_TSP = 4.2      # granulated white sugar
ADY_G_PER_TSP = 3.111     # active dry yeast (1 packet = 7g = 2.25 tsp)

def tsp_to_grams(tsp, g_per_tsp):
    return tsp * g_per_tsp

def score_against_reference(fvi_new, fvi_reference, reference_score=95.0,
                           cap_at_100=True):
    """
    Turn a raw FVI into a 0-100 "yeast health" gauge, anchored to a known
    run. 0 = no measurable pressure rise at all (dead/inactive yeast) --
    that's a real physical floor, not an arbitrary choice. The reference
    run defines the other end of the scale.

    Because the score is a straight ratio, if yeast_g and sugar_g are held
    IDENTICAL between the reference run and the new run, they cancel out of
    the ratio completely -- the mass-normalization terms only matter once
    you start varying the recipe between tests. Until then, think of this
    as purely a curve-shape-and-magnitude comparison.
    """
    s = reference_score * (fvi_new / fvi_reference)
    return min(s, 100.0) if cap_at_100 else s

```

```
if __name__ == "__main__":

    fit = fit_pressure_curve("input.csv") # Change the file name depending on the input
    file you would like to use

    print("Curve fit:", {k: round(v, 4) if isinstance(v, float) else v
                          for k, v in fit.items()})

    # >>> This run's actual recipe: 1/4 tsp active dry yeast, 1/2 tsp sugar
    YEAST_G = tsp_to_grams(0.25, ADY_G_PER_TSP)    # ~0.778 g
    SUGAR_G = tsp_to_grams(0.5, SUGAR_G_PER_TSP)   # ~2.1 g

    result = fermentation_vigor_index(fit, yeast_g=YEAST_G, sugar_g=SUGAR_G)
    print(f"\nyeast_g={YEAST_G:.3f}    sugar_g={SUGAR_G:.3f}")
    print("Fermentation Vigor Index:")
    for k, v in result.items():
        print(f"    {k}: {round(v, 4)}")

    # This run is the calibration anchor: user-assessed as 95/100.
    FVI_REFERENCE = 11.604056944342746
    REFERENCE_SCORE = 95.0

    # print(result['FVI'])

    print(f"\nYeast health score (this run): "
          f"{score_against_reference(result['FVI'], FVI_REFERENCE,
REFERENCE_SCORE):.1f} / 100")
```

Appendix K: Baseline Filtered Data (Small Font)

true_time,timestamp,uptime_ms,pressure_hPa,pressure_hPa_filtered,ds18b20_tempC,ds18b20_tempC_corrected,ds18b20_tempC_filtered,thermistor_mV,thermistor_tempC_calc,thermistor_tempC_filtered,lps33hw_tempC_avg_temp_filtered,temp_norm,pressure_norm,fermentation_activity_index,pressure_adjusted_thermistor,pressure_adjusted_ds18b20

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